

Five Facts About (Rental) Prices*

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Abstract

Rent is the largest component of the Consumer Price Index, yet how landlords adjust contract rents remains poorly understood. We uncover five stylized facts using 25 years of comprehensive administrative data from Berkeley, California. First, rents bunch at and just below multiples of \$100, consistent with efforts to exploit tenants' left-digit bias. We find that the degree of bunching is highly correlated with proxies for landlord sophistication, suggesting that these patterns reflect a combination of misoptimization and market power. Second, rent-setting displays strong nominal rigidity at tenant turnover: roughly one-fifth of units that change hands within two years show no price change, despite clear economic incentives to adjust and landlords already incurring the fixed cost of listing the unit. Third, rent adjustments are highly state-dependent: the frequency and size of both price increases and decreases move with market conditions, in contrast to most non-housing CPI components. Moreover, when inflation or deflation pushes rent growth away from typical trends, landlords engage in more round-number bunching and less left-digit pricing, revealing a novel cognitive cost of inflation. Fourth, rental prices exhibit pronounced seasonality, with summer lease starts commanding premiums of roughly 4 percent. Fifth, the sensitivity of rent adjustments to the business cycle varies by landlord size, with larger owners adjusting rents more flexibly than smaller ones. We replicate many of these patterns in nationally representative survey data. Together, these findings provide new moments for calibrating sticky-price models and highlight the role of cognitive frictions and firm heterogeneity in shaping inflation dynamics.

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1 Introduction

Housing is the largest component of the cost of living for most U.S. households, making shelter prices central to understanding household budgets and economic well-being. Shelter accounts for 35% of total expenditures in the Consumer Price Index (CPI), a larger share than all goods combined. These costs also play a key role in shaping aggregate inflation dynamics over the business cycle. Figure 1 plots key CPI components over the past 35 years. The black line shows the rental price index, while the pink line shows the CPI excluding shelter, food, and energy. For two decades, the non-shelter, sticky-price CPI remained relatively stable, fluctuating between 1% and 3% each quarter.

In contrast, the rental component of the CPI covaries strongly with the business cycle. Rental CPI fell by 4 p.p. during the Great Recession, before rebounding back to around 4% annually during the recovery. During the initial COVID recession, the non-shelter, sticky-price CPI was flat, while rental prices fell by 2 p.p. Even in the burst of inflation that began in 2021, the non-shelter components of CPI only peaked at around 6%. In contrast, rental price inflation peaked at nearly 9%.

Despite the central role of rental housing in shaping both aggregate inflation and household budgets, price-setting behavior in rental markets remains poorly understood. In contrast to the wealth of data on home sales, rental prices are not systematically collected, aside from surveys conducted by statistical agencies to construct the CPI. However, home prices capture asset values rather than just the flow cost of shelter that drives cost-of-living pressures and inflation dynamics.

Much of what we know about price setting in goods markets comes from retail scanner data, which links transactions to clearly defined products. No equivalent dataset exists for rental housing. Recent studies have used advertised rents from multiple listing services or platforms like Craigslist, but these data cover only selected segments of the market, are limited to recent years, and track posted, rather than transacted, rents. The composition of listings also shifts over time, complicating inference about broader market dynamics.

We use a unique administrative dataset covering the universe of contract rents in Berkeley, California, a city located in the San Francisco Bay Area and part of one of the nation’s most important rental markets. This dataset allows us to provide new evidence on rental price setting over the last 25 years. Our dataset allows us to follow the contract rents for new tenants at the unit level. Our setting is ideal for several reasons. First, Berkeley has experienced significant variation in rental prices over the last few decades. Thus, we can study rental price setting in regular times, as well as in periods when rental prices are falling

or rising rapidly. Our sample period includes multiple years when new tenant rents fell by more than 3%, as well as multiple years when rents increased by more than 10%. As a result, we avoid having to use only periods of moderate inflation to understand dynamics in high- or low-inflation periods (Nakamura et al., 2018; Montag and Villar, 2023; Gagliardone et al., 2025).

Second, the unit-level detail in our administrative dataset allows us to control carefully for housing quality when analyzing price setting and rental inflation. This is particularly important given the substantial rise in average housing quality over the past 30 years, which has widened the gap between raw rent measures and quality-adjusted housing costs (Molloy, 2024).

Finally, the institutional context of our setting closely resembles a Calvo (1983)-style pricing friction. In Berkeley, rents are tightly regulated once a tenant occupies a unit, and tenant protections are strong. However, landlords are free to set rents without restriction when a tenant vacates. While Calvo pricing is often used as a modeling device for price stickiness, few real-world settings so closely mirror its assumptions. Our data allow us to observe how landlords set prices under these frictions, offering novel empirical moments for calibrating Calvo models. Moreover, because landlords cannot adjust rents once a tenant is in place, tenancies often last many years, and the Berkeley rental market is characterized by strong average growth in new tenants’ rent, landlords have strong incentives to set prices optimally when new tenants enter a unit. Throughout our analysis of the Berkeley dataset, we only study rents for new tenants. Thus, none of our baseline estimates include rents that are subject to price controls.

Although our primary analysis uses granular administrative data from Berkeley, California, we reproduce many of the key patterns in a nationally representative survey of U.S. rental units. This alignment indicates that the mechanisms we identify extend well beyond the San Francisco Bay Area. At the same time, because Berkeley has experienced greater cyclical and seasonal variation than most U.S. rental markets, it provides a rich setting to study how aggregate shocks influence rent-setting behavior.

We use our dataset to document five new facts about the dynamics of rental housing prices. First, we provide novel evidence that landlords rely on heuristics when setting rents. The distribution of new contract rents shows substantial bunching: 35% are exact multiples of \$100, and nearly 60% are multiples of \$50. These estimates of bunching are substantially larger than previous estimates from the U.S., which used survey data from the 1970s and early 1980s (Genesove, 2003). However, we also find much larger estimates using modern updates

of the same survey dataset used in [Genesove \(2003\)](#) for the broader U.S. rental market, suggesting that increased threshold pricing is a feature of rental markets more broadly, and not specific to Berkeley, CA.

We also find pronounced left-digit pricing, with a sharp spike at prices exactly \$5 below round numbers. There are 39.5 times as many rents set \$1–\$9 below a \$100 multiple as above. Bunching and left-digit pricing vary systematically with proxies for landlord sophistication: large, corporate landlords rely less on round-number pricing but make greater use of strategic left-digit pricing. These patterns provide evidence of widespread pricing frictions and point to a role for tenants’ left-digit bias. The fact that less sophisticated landlords make apparent pricing mistakes yet still fill units suggests that market power allows for substantial heterogeneity in pricing behavior. Our results also suggest behavioral frictions on the seller side are a crucial determinant of pricing, even in a very high-stakes setting where sellers may be unable to update prices for a decade or more.

Second, many landlords do not update rents when a new tenant enters. The distribution of rent changes shows a pronounced spike at exactly \$0: 13% of tenant spells begin with the same rent as the previous initial contract for that unit. The likelihood of a zero rent change strongly correlates with the time between leases, a proxy for the distance from the incumbent price to the ideal reset price. One-quarter of rents that reset within one year remain unchanged, compared to just 1% after an eight-year gap. We also find evidence of downward nominal rent rigidity: there is a large mass of rent changes just above \$0 and a missing mass just below. These patterns surprisingly arise in a setting where (1) average rent growth is strong, (2) a new landlord-tenant match eliminates relational contracting concerns, and (3) the landlord is already incurring the menu costs of vacancy posting. Our first two facts point to behavioral frictions as central to understanding rent-setting behavior.

Third, we examine how rental prices respond to business cycle fluctuations. [Nakamura and Steinsson \(2008\)](#) show that for the non-shelter components of the CPI, the frequency of price increases rises with inflation. In contrast, the frequency of price decreases and the size of both price increases and decreases remain stable—patterns consistent with a benchmark menu-cost model. However, we find strikingly different behavior for the shelter component. We first document that the number of new tenant leases does not respond to changes in rental prices, consistent with a [Calvo \(1983\)](#) framework in which tenant turnover is driven by exogenous mobility shocks uncorrelated with macroeconomic conditions.

However, conditional on a lease reset, rental pricing dynamics display significant state dependence. The frequency of rent increases rises sharply with overall rent growth, while

the frequency of rent decreases and zero price changes decline. The magnitude of both price increases and decreases also covaries strongly with rent growth. As a result, nominal rigidity is asymmetric over the cycle: the excess mass just above zero and the missing mass just below zero in the rent change distribution is present in expansions, but not in years surrounding recessions, when rents are falling. Despite patterns consistent with strong nominal frictions, landlords do cut rents during downturns. We find very similar results in the broader sample of U.S. rental units. These new moments sharpen our understanding of rent-setting behavior, revealing that pricing is highly state-dependent and shifts over the business cycle.

We also show that behavioral pricing frictions are state-dependent. As annual rent inflation deviates further from its median level, landlords rely more heavily on heuristic pricing: they are more likely to set rents at multiples of \$50 or \$100 and less likely to engage in left-digit pricing. We interpret this as evidence that unusual nominal changes impose cognitive costs that impair optimization precision. We validate this mechanism using variation in the time between rental contracts: longer gaps, which lead to larger ideal price adjustments, are associated with more threshold pricing and less left-digit pricing. These results point to a novel cost of inflation omitted from standard New Keynesian models: as inflation increases, the complexity of price setting rises, leading to less precise pricing decisions.

Fourth, we show that predictable seasonal variation in rental demand has large effects on prices. In Berkeley, nearly 60% of leases begin during the summer months of June, July, or August. Leases that start outside of this peak period are more likely to feature rent decreases or zero price changes and less likely to involve large rent hikes of \$250 or more. On average, off-peak rentals are discounted by 4.5%. While Berkeley has an unusually large seasonal cycle tied to the academic calendar, we find very similar results in the broader U.S. renter sample. Despite the smaller seasonal cycle in nationally representative data, we estimate nearly identical elasticities of rental prices to seasonal variation in leases. [Ngai and Tenreiro \(2014\)](#) document similarly strong seasonality in the home purchase market and argue that it is inconsistent with standard housing models, suggesting instead that thick-market effects amplify seasonal moving patterns. The fact that we find comparable seasonality in the rental market, despite the shorter duration of rental contracts and weaker incentives to maximize match quality, suggests that other forces may be driving seasonality across both rental and owner-occupied housing markets. Our third and fourth facts document how prices adjust to cyclical and seasonal fluctuations, and demonstrate that behavioral pricing patterns are themselves sensitive to aggregate economic conditions.

Finally, we document striking heterogeneity in price-setting behavior by landlord type.

In addition to showing less evidence of behavioral pricing biases and being more likely to exploit the left-digit bias of tenants, larger landlords are less likely to leave rents unchanged between tenant spells. They are also more likely to offer rent cuts and, when raising rents, tend to do so by larger amounts. Much of this heterogeneity reflects differences in responsiveness to market conditions. Large landlords adjust prices more aggressively over the business cycle: they are more likely to cut rents during recessions and raise rents, and by larger amounts, during expansions. They also respond more to seasonal fluctuations, offering steeper discounts for off-peak leases.

In contrast, smaller landlords exhibit stronger state dependence in heuristic pricing. As inflation deviates from typical levels, they are more likely to rely on round-number pricing and less likely to use left-digit strategies, consistent with higher cognitive costs and lower pricing precision. These findings suggest that larger landlords are more sophisticated and responsive to economic conditions, while smaller landlords face greater behavioral frictions. Our fifth fact highlights how firm heterogeneity shapes both inflation dynamics and the responsiveness of rents to shocks. These findings also carry important welfare implications, particularly given ongoing debates over the growing role of large, corporate landlords in the rental market.

1.1 Literature

Our work contributes to three main strands of the literature. First, we build on foundational research documenting price-setting behavior for non-housing components of the CPI. Studies such as [Bils and Klenow \(2004\)](#), [Nakamura and Steinsson \(2008\)](#), [Klenow and Kryvtsov \(2008\)](#), and [Klenow and Malin \(2010\)](#) establish empirical facts that have shaped how economists model pricing frictions. In contrast, we present new statistics on shelter prices—the largest component of household expenditure and a central factor in recent inflation dynamics. Institutionally, our setting closely resembles the staggered pricing friction introduced by [Calvo \(1983\)](#). While Calvo pricing is often invoked for tractability, few real-world environments closely match its structure. [Bils and Klenow \(2004\)](#) and [Nakamura and Steinsson \(2008\)](#) show that many observed facts are more consistent with menu-cost models. In contrast, we offer new moments that reflect firm behavior under Calvo-type frictions. We also document state-dependent pricing in the rental market, contributing to recent work on how price-setting behavior evolves in high- and low-inflation environments ([Golosov and Lucas Jr., 2007](#); [Nakamura et al., 2018](#); [Montag and Villar, 2023](#); [Gagliardone et al., 2025](#)). We document the effects of seasonality in the rental market, contributing to a broader literature

that leverages seasonal patterns to study economic behavior (Barsky and Miron, 1989). Our work on how rents vary over the business cycle can also inform recent papers examining the optimal policy response to housing price inflation (Bianchi et al., 2024).

Second, we contribute to the literature on rental price dynamics. Prior work has documented nominal rent stickiness using household surveys (Genesove, 2003; Hoffmann and Kurz-Kim, 2006; Shimizu et al., 2010; Aysoy et al., 2014; Gallin and Verbrugge, 2019), but such data often suffer from recall error and limited coverage. In contrast, we leverage an administrative dataset based on legally reported contract rents submitted by landlords. We also study rent setting over decades, which allows us to analyze how rental-price setting is affected by cyclical variation. We provide evidence of pronounced state dependence in rental price setting that is novel to the literature. We also show that nominal rigidity and heuristic pricing are endogenous responses to inflation, the timing of lease resets, and landlord characteristics. Our results highlight behavioral frictions as key drivers of sticky rent-setting behavior.

While our analysis focuses on rents for new tenants, we complement recent work by Adams et al. (2024), who emphasizes the differences between pricing for new and continuing tenants. Calder-Wang and Kim (2024) study how the adoption of AI pricing tools affects asking rents using a large sample of multi-unit buildings. Our work also relates to Genesove and Mayer (2001), who document how behavioral frictions influence home sale and purchase decisions. Baker (2024) uses the same Berkeley dataset as we do to study pass-through of property taxes to rents, and finds results consistent with non-standard, behavioral pricing patterns.

Finally, our findings relate to a growing literature on behavioral frictions in firm decision-making (Matějka, 2016; DellaVigna and Gentzkow, 2019; Stevens, 2020; Strulov-Shlain, 2022, 2024). We show that behavioral considerations—such as heuristic pricing and inattention—are first-order in the rental housing market, and that their influence varies systematically with the identity of the seller and the business cycle. Consistent with Dube et al. (2018), who interpret round-number bunching in wages as a joint outcome of firm misoptimization and market power, we document pronounced bunching in the rent distribution and interpret it as reflecting both landlord misoptimization and pricing power.

2 Setting & Data

2.1 Berkeley Rent Registry

We use administrative data from the Berkeley Rent Board covering 1980–2022. The Rent Board collects detailed information on rental units subject to rent control or eviction protections, including unit characteristics, initial rent, and lease start dates. Our primary sample focuses on new-tenant rents from 1999 onward, after the adoption of vacancy decontrol under the Costa-Hawkins Rental Housing Act of 1995. These rents are freely set by landlords and not subject to rent control restrictions.

We exclude the small number of units with more than five bedrooms, which tend to either be very large homes for students (frat houses) or are the very richest segment of the rental market. The dataset covers 19,000–26,000 units annually, with the number increasing as more units were built and/or registered with the Rent Board.

We link Rent Board records to property tax data from the Alameda County Assessor’s Office, which provides owner names and mailing addresses. Corporate landlords are identified by terms like LLC, Corp, or Company in the owner name. We group properties by owner mailing address to construct a proxy for landlord size.

2.2 Berkeley Rental Market

Berkeley, California, is part of the San Francisco Bay Area, one of the most competitive rental markets in the U.S., with persistently low vacancy rates around 5%.¹ Figure 2, Panel (A), plots a repeat-rent index for new-tenant rents in Berkeley since 1999. The index includes unit fixed effects, capturing within-unit price changes over time and adjusting for time-invariant quality. Quality-adjusted rents rose by over 80% during our sample period, though growth was uneven. Rents declined after the 2001 and 2008 recessions and were flat from 2018 to 2021.

Panel (B) shows the estimated hazard rate of new rental contracts.² The hazard declines with incumbent tenure, with sharp spikes at 12-month intervals, reflecting standard lease lengths. While the median time between leases is two years, over 25% of leases last more than three years, and some exceed ten. This long right tail, combined with rising rents and limited ability to adjust prices within leases, creates strong incentives for landlords to set

¹[Housing Vacancies and Homeownership \(CPS/HVS\)](#), U.S. Census Bureau.

²We truncate the graph at 100 months; about 4% of new leases begin more than 100 months after the previous one.

rents carefully when units turn over.

Our setting closely mirrors the pricing friction introduced by [Calvo \(1983\)](#). Landlords can freely set rents when a tenant exits, but cannot raise rents during an ongoing tenancy in a rent-controlled unit.³ Strong tenant protections in Berkeley further limit evictions and, in some cases, restrict rent increases following an eviction.⁴ Because landlords reset prices precisely when they must already re-list the unit and match with a new tenant, frictions such as menu costs or buyer-seller relationships are minimal. This makes our setting a clean environment in which to study price-setting behavior under Calvo frictions.

2.3 American Housing Survey

We use the American Housing Survey (AHS) to test whether our findings generalize to the broader American rental market. The AHS is a longitudinal panel survey conducted by the Census Bureau that tracks individual housing units over time.⁵ We focus on the 1985 panel, which follows the same houses every other year from 1985 to 2013.⁶

Despite its broader scope, the AHS has several critical limitations that led us to use the Berkeley registry as our primary data source. First, the AHS does not capture all new contracts, even for sampled units. Second, we cannot differentiate between new tenant contracts and continuing tenant contracts, a distinction crucial for analysis as these contract types may differ systematically ([Adams et al., 2024](#)). Third, the AHS provides less business cycle variation in rents because the aggregate U.S. rental market is less volatile than the San Francisco Bay Area, observations occur only biennially, and the AHS lacks detailed geographic information to precisely locate properties. Fourth, the AHS contains no data on landlord characteristics, which our Berkeley analysis reveals as a significant driver of pricing behavior. Finally, as a survey completed by tenants, the AHS may suffer from non-response bias and imprecise recall of rental contract details—particularly problematic for our research, which depends on precise dollar amounts.

To construct our main sample, we use AHS data from 1985 to 2013, focusing exclusively on rental properties and excluding mobile homes to enhance comparability with the Berkeley

³The main deviation from the Calvo model is that the reset hazard is downward sloping rather than constant, likely due to tenant-unit match effects. We also show in Section 3.3 that the number of new tenant contracts does not vary with the business cycle.

⁴In many cases, rent control laws prevent landlords from raising rents after an eviction.

⁵New homes are added to the sample as they are constructed.

⁶We do not use data from the 2015 panel onwards because many crucial variables are either suppressed for confidentiality or rounded, making it impossible to study questions related to threshold pricing, left-digit pricing, or sticky rents.

sample. We restrict our analysis to metropolitan areas, including center cities and urban suburbs, while excluding households in rural areas. We link units longitudinally to measure changes in rents over time. To better approximate our new tenant sample, we only include observations where the current household differs from the one interviewed in the previous survey wave. All observations are weighted using the Census Bureau’s sampling weights.

3 Empirical Results

3.1 Fact 1: Heuristic Pricing

We first provide evidence that landlords use heuristics when setting rents, leading to substantial bunching in the rental price distribution. Figure 3 shows the distribution of rents based on their distance from the nearest multiple of 100. Three patterns stand out. First, a large share of rents bunch exactly at multiples of 100. In our main sample, 35% of observations fall at these round numbers—a figure similar to the share of wages ending in round numbers reported in the CPS ([Dube et al., 2018](#)).

Second, we observe a smaller but still significant spike at rents that are multiples of 50 but not 100. In our baseline sample, about 24% of observations fall into this category. We also find smaller bunching at finer increments—multiples of 25, 10, and 5. In the full sample, 97% of initial rents are at multiples of 5, and 99% are whole numbers. These bunching rates are substantially higher than those documented in earlier survey data. [Genesove \(2003\)](#) finds that 7–11% of rents were set at multiples of 100, and 8–12% at multiples of 50 (but not 100) in the 1970s and early 1980s. In contrast, our estimates are more than three times higher for multiples of 100 and about twice as high for multiples of 50.

Third, the rental price distribution shows clear evidence of left-digit pricing ([Anderson and Simester, 2003](#); [Lacetera et al., 2012](#); [Strulov-Shlain, 2022, 2024](#)). There is a notable spike at exactly \$5 below a multiple of \$100, with this mass equal to approximately 42% of the spike at the corresponding multiple of \$100. This suggests that landlords believe left-digit bias significantly influences tenant behavior.

More broadly, we observe excess mass just below round numbers and missing mass just above. There are roughly 39.5 times as many observations in the interval $x \in [-9, 0)$ as in $x \in (0, 9]$ (where x is the distance from the nearest multiple of \$100). Even excluding the large spike at \$5 below multiples of \$100, there are over twice as many rents in $(-25, 0)$ as

in $(0, 25)$.⁷ To our knowledge, this is the first paper to document left-digit pricing in the rental housing market.

We find similar evidence of heuristic pricing in the AHS sample of U.S. renter households. Appendix Figure A5, Panel (A) reveals substantial bunching in the nominal rent distribution, though slightly less pronounced than in the Berkeley sample. In the AHS, approximately 23% of rents are set at multiples of \$100, and 41% at multiples of \$50. This prevalence exceeds that reported by [Genesove \(2003\)](#) using the same dataset for an earlier period, suggesting that bunching is a pervasive feature of modern rental markets broadly, not specific to Berkeley.

The differences likely reflect that bunching at round multiples becomes more desirable as nominal prices increase. Appendix Figure A5, Panel (B), restricted to 1999-2013 (more closely aligned with our Berkeley sample timeframe), shows increased bunching: 29% of rents at \$100 multiples and 48% at \$50 multiples. Overall, we find strong evidence of threshold pricing comparable to our results for Berkeley alone, with variations likely explained by Berkeley’s higher rent levels and the later sample period. Our results suggest that threshold pricing has increased substantially over time and that rising nominal rent levels may further amplify round-number pricing.

In contrast, we observe substantially less left-digit pricing in the broader AHS sample compared to the Berkeley administrative data. The AHS still exhibits a clear spike at prices exactly \$5 below each \$100 threshold, but that spike is only 17% as large as the spike at the round \$100 itself, versus 42% in Berkeley. These differences may stem from the AHS capturing both continuing and new-tenant rents, which demonstrate distinct dynamics ([Adams et al., 2024](#)), and from the superior precision of our administrative data, particularly if respondents struggle to recall non-round figures.

This disparity likely does not reflect Berkeley-specific characteristics, such as its substantial student population. Appendix Figure A1 maps left-digit bunching across neighborhoods. Although we detect considerable bunching in student-dominated areas near downtown and U.C. Berkeley’s campus, we also find substantial bunching in neighborhoods with minimal student presence, including southern areas bordering Oakland, West Berkeley near the marina, and the affluent Berkeley hills. By design, our dataset excludes much of the housing for undergraduate students—campus dormitories, large shared houses, student co-ops, and Greek residences. Notably, only one census tract demonstrates less bunching than observed in the AHS. Our analysis underscores the value of contract-level administrative datasets and illuminates the need for additional research utilizing similar rent registries to assess left-digit

⁷For comparability, we exclude observations exactly \$5 above multiples of \$100.

bunching patterns in the broader rental market.

Figure 4 uses the Berkeley dataset to show that landlord characteristics strongly predict pricing behavior. Panel (A) splits the sample by landlord size, defining large landlords as those owning more than 75 units in Berkeley. Small landlords exhibit significantly more bunching: over 40% of their rents fall at multiples of \$100, compared to less than 30% for large landlords. They also bunch more at multiples of \$50 (but not \$100), though the difference is smaller. In total, about two-thirds of rents set by small landlords are multiples of \$50, versus just half for large landlords.

In contrast, left-digit pricing is far more common among large landlords than small ones. Over one-quarter of rents set by large landlords are exactly \$5 below a multiple of \$100, whereas the share for small landlords is below one-tenth. Figure 4, Panel (B), shows a similar pattern when we classify units by owner type: individual landlords bunch more at \$50 and \$100 grid points but are much less likely to use left-digit pricing than corporate owners.

Figure 5 plots binscatters of bunching probabilities against landlord size. Across all three outcomes—bunching at \$100, bunching at \$50, and left-digit pricing—we find a strong linear relationship between the probability of bunching and the number of units the landlord owns in Berkeley (in logs).⁸ Among the smallest landlords, nearly 50% of rents lie at multiples of \$100 and left-digit pricing is almost absent. For the largest landlords, only 25% of rents are at \$100 grid points, while more than 30% employ left-digit pricing.

These patterns confirm that heuristic pricing materially shapes rent setting. Grid pricing at \$100 and \$50 is far more prevalent in our data than in earlier U.S. evidence on rental markets (Genesove, 2003) and is comparable in magnitude to round-number bunching in wages (Dube et al., 2018). We also document a pronounced spike exactly \$5 below round hundreds and excess mass just below, with missing mass just above, multiples of \$100—clear evidence of left-digit pricing.

Finally, the degree of heuristic pricing varies systematically with landlord characteristics. Large, corporate landlords bunch less at grid points but engage more in left-digit pricing. We interpret these facts as reflecting a mix of landlord misoptimization, market power, and tenant left-digit bias (Dube et al., 2018). Because competition in rental markets is imperfect, landlords can make pricing errors while still filling their units. The greater reliance on grid pricing by less-sophisticated landlords (proxied by smaller portfolios or individual ownership) points to behavioral frictions on the landlord side, not solely tenant preferences, as a key

⁸All estimates are residualized on census-tract-by-year fixed effects.

driver of bunching.

3.2 Fact 2: Nominal Rent Rigidity

Our second fact is that rents exhibit strong evidence of nominal rigidity. Figure 6, Panel (A) plots the distribution of rent changes for new tenants at each nominal dollar value.⁹ A large spike appears at exactly zero: in our main sample, 13% of new-tenant leases have rents that are identical to those charged to the previous tenant. Strikingly, this is the same share reported by Genesove (2003), despite their study using a different dataset and a sample period two decades earlier. Both Genesove (2003) and Aysoy et al. (2014) argue that a substantial portion of nominal rent rigidity can be explained by grid pricing. However, since we observe the same level of rigidity despite finding a much larger degree of grid pricing, our results suggest that grid pricing alone cannot provide a causal explanation for nominal rent stickiness.

Figure 7, Panel (A) shows that nominal rigidity is responsive to economic incentives. The cost of not adjusting rent increases with the time between leases, as market prices drift away from the previously set rent. Consistent with this, the probability that landlords leave rent unchanged declines sharply as the gap between tenants grows—from nearly 30% for short gaps to nearly zero after longer intervals. This pattern suggests that landlords face a cost to reoptimizing, which they are more willing to pay as the potential pricing error grows.

Rent changes are also asymmetric. Figure 6, Panel (A) shows excess mass just above zero in the rent change distribution. For example, there are roughly 2.5 times as many rent increases in the (0, 50] USD range as there are rent decreases in the [-50,0) USD range. Figure 7, Panel (B) shows the kernel density of log rent changes after excluding zeros. Two features stand out: (1) rent increases are far more common than decreases, and (2) rent changes bunch just above zero, with many small increases but very few small cuts.

Figure A6 panel (A) plots the distribution of two-year rent changes in the AHS. Zero changes are harder to spot in this survey because rents are observed only biennially, so the shortest leases—the ones that create the most bunching—go unrecorded.¹⁰ Even so, panel (A) reveals a nearly comparable share of unchanged rents: roughly 8 percent after two years, versus 13 percent for new-tenant rents in Berkeley.¹¹

⁹Rent changes are rounded to the nearest dollar.

¹⁰Genesove (2003), using an earlier annual AHS, documented bunching identical to that in Berkeley, reinforcing the external validity of our Berkeley results.

¹¹The calculation uses only units with a household turnover between waves, so it is not driven by continuing tenants.

Panel (B) plots the kernel density of non-zero two-year rent changes in the AHS and also finds evidence of downward-nominal rigidity: missing mass just below zero and excess mass just above it. Thus the broader AHS corroborates the Berkeley patterns, while our transaction-level registry provides the granularity needed to detect these pricing frictions with greater precision.

These patterns closely resemble those documented in wage-setting, where small increases are common and small decreases are rare (Hazell and Taksa, 2025). In the labor market, this asymmetry is often attributed to workers’ aversion to wage cuts (Davis and Krolkowski, 2025). But in the rental market, that mechanism is absent—suggesting that downward nominal rigidity may also arise from firm-side frictions, such as cognitive constraints or inattention. Given that downward nominal wage rigidity is widely viewed as a key friction in labor markets, our findings raise the possibility that similar pricing constraints may play an important role in shaping housing market dynamics.

3.3 Fact 3: Prices Respond to the Business Cycle

We next examine how housing markets respond to cyclical variation. Nakamura and Steinsson (2008) show that, for non-housing components of the CPI, the frequency of price increases covaries with inflation. In contrast, the magnitude of price increases and both the frequency and magnitude of price decreases are largely unresponsive. These patterns are consistent with benchmark menu-cost models.

In the housing component of the CPI, we find markedly different dynamics. Figure 8, Panel (A), plots the total number of new leases each year. Gray shading marks years in which nominal rents declined—following the 2001 and 2007 recessions, as well as in 2020. There is no clear cyclical pattern in the total volume of rental contracts, suggesting that the overall frequency of moves does not respond to the business cycle. This pattern aligns with a Calvo-style model in which landlords’ ability to reset rents is exogenous and is not itself state-dependent. Table 1, Column (1), corroborates this interpretation by showing that the number of new leases is uncorrelated with changes in the new-tenant rent price index.

While the total number of new leases appears acyclical, we find strong evidence that the composition of rent changes is highly responsive to shifts in rental demand. Figure 8, Panel (B), shows that in periods of weak demand (shown by gray shaded bars), the share of rent increases drops sharply, while rent cuts become more common. Following the 2001 recession, rent decreases became nearly as frequent as rent increases for new tenants. Similar, though somewhat less extreme, patterns emerged during the Great Recession and the late-2010s

rental slowdown, which culminated in the 2020 COVID recession. In both cases, more than 20% of new leases included rent reductions in at least one year.

We also observe a similar, but less pronounced and more delayed, pattern in the share of rents that remain unchanged between tenants. Following the 2001 and 2008 recessions, the fraction of new leases with zero price changes reached a local maximum of around 20%. These peaks were smaller than those observed for rent decreases and occurred later: the peak in unchanged rents followed the peak in rent cuts by two years after the 2001 recession and by one year after the 2008 recession. Table 1, Columns (2)–(4), reports the time-series relationship between the number of rent increases, decreases, and unchanged rents and the change in the rental price index. All three outcomes are strongly correlated with changes in the index and are highly statistically significant. Overall, we find compelling evidence that the frequency of rent increases, decreases, and unchanged rents is closely tied to the business cycle. This stands in contrast to the patterns documented by [Nakamura and Steinsson \(2008\)](#) for non-housing prices, where only the frequency of price increases moved systematically with inflation.

At first glance, these results appear to contradict the evidence presented in Figures 6 and 7, which suggest that landlords face a constraint that prevents rent reductions. However, Figure 9 shows that downward nominal rent rigidity is state-dependent. This figure plots the kernel density of rent changes (excluding zeros) separately for economic expansions and contractions. The asymmetry near zero, a concentration of mass just above zero, and missing mass just below is only present during expansions. In contractions, this asymmetry disappears. Appendix Figure A6, Panel (C), shows similar patterns in the broader AHS sample of U.S. renters.¹² Together, these results suggest that landlords are reluctant to cut rents in strong markets but do adjust downward in response to adverse demand shocks.

This form of state-dependent rigidity starkly contrasts with evidence of downward nominal wage rigidity, where constraints appear binding, and firms do not cut wages, even in downturns ([Hazell and Taksa, 2025](#)). Our results imply that downward nominal rent rigidity is driven by behavioral frictions that vary with economic conditions and, thus, does not represent a binding constraint on landlords during downturns. Consequently, rent rigidity is unlikely to play a central role in amplifying business cycle fluctuations.

Figure 10 shows that the magnitude of rent changes also varies systematically with the business cycle. The figure plots the median size of rent increases and decreases over time, separately for new leases following a turnover within one year or less and for all new leases.

¹²We cannot show contractions in the rent index because we do not see many years of negative price growth in the broader U.S. rental market.

Panel (A) illustrates that, for recently turned-over units, the magnitude of rent changes is closely tied to cyclical conditions. The median size of rent increases rises during expansions and falls during contractions, while the opposite pattern is observed for rent decreases. Table 2, Columns (1) and (2) confirm these patterns. In addition, Table 2 shows that the sensitivity of rent increases to changes in the rent index is roughly twice as large as that of rent decreases.

Panel (B) shows that these relationships persist when examining all new leases, although the procyclicality of rent increases becomes even more pronounced. Table 2, Columns (3) and (4) support this finding: the relationship between the size of rent increases and the rent index nearly doubles relative to the subsample of recently turned-over leases. By contrast, the sensitivity of rent decreases remains virtually unchanged. As a result, in the full sample, the responsiveness of rent increases to the business cycle is approximately 3.5 times greater than that of rent decreases.

Although aggregate renter mobility remains flat over the business cycle, the mix of movers or units could still shift. Appendix Table A1 addresses this concern by estimating a regression at the contract level with unit fixed effects. Thus, the estimates compare rent-setting for the same rental unit at different stages of the business cycle.

The within-unit results echo the aggregate patterns. Higher growth in the rent price index raises the probability of a rent increase and lowers the likelihood of either a rent cut or no change. The shares of increases and cuts move more with the cycle than the share of zero changes. To gauge the effect on magnitudes, Columns (4) and (5) report the probability of a large adjustment—defined as an increase above \$250 or a cut of \$100 or more. Large increases rise steeply with the rent index, whereas large cuts fall sharply.

Appendix Table A5 repeats the within-unit regression in the AHS sample. The coefficients closely track those from Berkeley. A one-percentage-point increase in the rent-price index produces slightly larger absolute effects on the probabilities of both a rent hike and a rent cut, while the probability of an unchanged rent remains identical across datasets. In Berkeley, large increases ($>\$250$) respond more to the cycle than large decreases ($<-\$100$); nationwide, the ordering reverses.¹³ The elasticity of log rent changes is also modestly higher in the AHS, likely because the national index blends new and continuing leases, whereas the Berkeley index covers only new leases and continuing-tenant rents are typically stickier (Adams et al., 2024). Overall, the near-parallel estimates indicate that the business-cycle pricing patterns documented for Berkeley generalize to the broader U.S. rental market.

¹³Because average rents are lower nationally, we define large changes as \$100 for increases and $<-\$50$ for decreases.

Finally, we show that the business cycle also influences the extent of heuristic pricing. Figure 11 reveals a strong, non-monotonic relationship between the prevalence of heuristic pricing and changes in the rent price index. Panels (A) and (B) show that bunching at round numbers—multiples of 50 and 100—is lowest when rent growth is near its median value (approximately 4.2% over the sample period, denoted by the dashed vertical line), and significantly higher when rents are either falling or rising sharply. In contrast, Panel (C) shows the opposite pattern for left-digit pricing: it is most prevalent during periods of moderate rent growth and declines during unusually large increases or decreases in rents. For all three outcomes, we estimate precisely measured, statistically significant differences in slope on either side of the median rent growth rate.

This non-monotonic relationship suggests that the magnitude of nominal rent changes shapes the cognitive constraints underlying heuristic pricing. During periods of typical rent growth, landlords appear able to optimize more precisely—avoiding coarse pricing grids and strategically exploiting left-digit biases. However, when rents fall or rise dramatically, the large nominal adjustments required may overwhelm landlords’ ability to fine-tune prices. As a result, we observe greater reliance on round-number pricing and a decline in left-digit pricing under such conditions.

We validate this interpretation using a separate source of variation in the optimal nominal reset price. When more time passes between new-tenant leases, the optimal rent drifts further from the incumbent contract. As a result, reset prices should increase with the time between rental contracts. We confirm this empirically in Appendix Figure A2, which shows that reset prices rise with time between leases, with an elasticity of approximately 0.15.¹⁴ Appendix Figure A3 shows that these larger price changes, induced by longer durations between leases, produce the same heuristic pricing patterns that we observe with business cycle variation. When the gap between the incumbent and reset price is larger, landlords exhibit more bunching at round-number prices: at multiples of 100 (Panel A) and 50 (Panel B), and less reliance on left-digit pricing (Panel C). This provides direct evidence for our proposed mechanism: larger price changes impose cognitive constraints that lead landlords to rely more heavily on heuristics, reducing the precision of their price-setting behavior.

These findings reveal a novel cost of inflation absent from standard New Keynesian models: large nominal price changes introduce cognitive frictions that reduce firms’ ability to optimize. We show that rent adjustments over the business cycle differ sharply from non-housing CPI components, aligning more with Calvo-style frictions than with menu-cost mod-

¹⁴This specification residualizes neighborhood-by-year fixed effects, thereby removing variation due to business cycle fluctuations or neighborhood-specific trends.

els. Behavioral pricing anomalies, such as threshold pricing, left-digit bias, and downward nominal rigidity, are state-dependent and vary systematically with aggregate conditions. Inflation thus affects not only price levels but also the precision of firms’ decision-making, shaping how shocks propagate through the economy.

3.4 Fact 4: Seasonal Demand and Pricing

Fourth, we document how seasonality shapes housing demand, pricing, and rent changes. Figure 12 displays the number of lease starts by month over the sample period. Two patterns emerge. First, most leases are signed during the summer months of June through August. Second, lease activity drops sharply between October and April. While Berkeley’s status as a university town likely contributes to the summer spike in demand, this pattern is not unique: nationally, 31.3% of leases begin between June and August, according to the BLS.¹⁵ The level of seasonality we estimate for the Berkeley rental market closely mirrors that of the U.S. home purchase market, as documented by [Ngai and Tenreyro \(2014\)](#).

The timing of lease start dates also affects rent levels and the distribution of price changes. Figure 13, Panel (A), shows the distribution of rent changes for peak-season leases (June–August) versus off-peak leases (September–May). Rent changes during the peak season are more likely to include large increases (greater than \$100), while off-peak leases more frequently involve smaller increases or rent cuts. Table 3 provides supporting evidence from an OLS regression estimating how seasonal timing affects rents within the same rental unit. Column (1) confirms a substantial seasonal increase in rental activity during the summer, with 148% more new leases in peak months. Columns (2) and (3) show that leases with no rent change are about 1 percentage point less likely during these months, while rent cuts are 4 percentage points less likely. Column (4) shows that large rent increases (exceeding \$250) are 2.6 percentage points more likely during peak periods. Column (6) indicates that leases signed during peak months are, on average, about 4.4% more expensive. However, Column (5) shows that the prevalence of left-digit pricing does not vary with seasonality. In sum, rental contracts started during summer months command higher rents, are less likely to include rent cuts, and are more likely to involve significant increases.

Figure 13, Panel (B), further illustrates the link between lease demand and pricing. The figure plots the estimated monthly rent premium (relative to January), based on a repeat-rent index, against the number of new leases started in each month. A clear positive relationship emerges: months with higher leasing activity are associated with higher rental

¹⁵“Housing Leases in the U.S. Rental Market,” 2022, BLS.

prices. We estimate an elasticity of new-tenant rents with respect to the volume of new leases of approximately 0.03.

Appendix Figure A7 replicates our seasonality estimates in the national AHS data. Panel (A) confirms a U.S. seasonal cycle, but one that is far milder than the Berkeley pattern. Panel (B) plots the monthly rent premia against log new-lease volume and again reveals a strong positive slope: months that generate more lease starts command higher rents. The estimated elasticity is 0.03. This is identical to the Berkeley value and shows that the same one-log-point increase in lease volume raises rents by roughly three basis points in both markets.

Appendix Table A6 repeats the within-unit seasonality specification in the AHS data. Panel (A) indicates that peak months nationwide bring only about one-quarter as many additional leases as Berkeley. Even with lower turnover, the pricing response remains significant, though smaller: peak months feature more rent hikes, fewer zeros, and fewer cuts, together with a 1.2 percent seasonal rent premium. Adjusting for the larger seasonal swing in lease volume observed in Berkeley suggests that, if the U.S. rental cycle were as pronounced, peak-month rents would rise by about 5%—close to the 4.4% premium we estimate in Berkeley. The parallel patterns confirm that the seasonal mechanisms identified in Berkeley operate nationally, and are not driven by the fact that the seasonal cycle is tied to the academic calendar or other features of the Berkeley market.

Our seasonality estimates parallel those of [Ngai and Tenreyro \(2014\)](#) in the home purchase market: the hot season lifts leasing volume by 148 percent and rents by 4.4 percent in Berkeley (about 5 percent nationally), almost matching their 146 percent surge in home sales and 4.6 percent rise in purchase prices. Rental contracts are far less durable and confer only modest match-specific surplus. Therefore, the pronounced seasonal rent premium cannot stem from the match effects central to their model. Because both markets exhibit virtually identical price–volume seasonality despite stark differences in match value, an additional mechanism must be operating alongside match effects to generate the observed patterns in both rental and home-purchase markets.

3.5 Fact 5: Heterogeneous Pricing Behavior by Landlord Type

Our final fact is that landlord identity is a strong predictor of how rents respond to the business cycle. As shown in Section 3.1, large and corporate landlords are more likely to exploit left-digit bias and less likely to set rents at coarse grid points such as multiples of \$50 or \$100. This pattern extends to rent changes as well. Table 4 estimates how rental price

characteristics vary with landlord size, controlling for neighborhood-by-year fixed effects. Column (1) shows that larger landlords exhibit less nominal rigidity: the probability of leaving rents unchanged between tenants is lower for larger landlords. Column (2) shows that they are also more likely to cut rents. The magnitudes in Columns (1) and (2) are similar and nearly offsetting, suggesting that larger landlords are less subject to the downward nominal rent rigidity observed during expansions (as documented in Section 3.2). These results underscore the importance of behavioral frictions or cognitive constraints that vary with landlord characteristics.

Column (3) of Table 4 shows that larger landlords are more likely to impose very large rent increases (greater than \$250), while Column (4) confirms that they raise rents by more on average between tenants. Appendix Tables A2 and A3 show consistent results when comparing corporate to individual landlords, and when defining large landlords as those owning more than 75 units in Berkeley.¹⁶

Table 5 shows that these average differences are largely driven by differential responses to the business cycle. The table reports regressions of rent change characteristics on landlord size interacted with terciles of average rent growth in Berkeley. Columns (1) and (2) show that larger landlords are more likely to raise rents during expansions and to cut rents during contractions. That is, the extensive margin of rent changes is more cyclical for large landlords. By construction, the difference between Columns (1) and (2) reflects the probability of a zero price change; while we omit this coefficient for readability, we find that larger landlords are less likely to leave rents unchanged, particularly in the middle tercile of rent growth. This again supports the conclusion that downward nominal rent rigidity is primarily present in expansions and that larger landlords are less subject to this friction.

Columns (3) and (4) of Table 5 show that differences in the intensive margin of price changes are also driven by differential responsiveness to the business cycle. Column (3) finds that larger landlords are more likely to impose very large rent increases—but only during periods of substantial rent growth. Consequently, Column (4) shows that the result that larger landlords raise rents more on average is entirely driven by their behavior in the top tercile of price growth. In sum, landlords of different sizes respond very differently to the business cycle on both the extensive and intensive margins.

We also find that large landlords are more responsive not only to volatility driven by

¹⁶Appendix Figure A4 presents the unconditional distribution of rent changes by landlord size quartile and corporate status. Some qualitative patterns differ when not conditioning on neighborhood-by-year fixed effects, suggesting that neighborhood characteristics and time trends are correlated with landlord traits and influence rent dynamics.

the business cycle, but also to seasonality in rental markets. Appendix Table A4 examines how within-unit responsiveness to seasonal cycles varies with landlord size. While we find limited evidence of heterogeneity by landlord size in the propensity to cut rents or give increases of larger than \$250, we find that larger landlords systematically raise rents more on average when leases are signed during the summer. The magnitude of our estimates suggests that moving from a mom-and-pop landlord with a single unit to a large landlord with 75 units nearly doubles the seasonal price premium associated with peak months relative to off-peak periods. This result is consistent with our broader evidence that larger landlords exhibit more sophisticated pricing behavior and are better able to adapt to both predictable patterns in demand, such as seasonality, and unpredictable shocks, such as those arising from the business cycle.

Table 5 also highlights that reliance on pricing heuristics varies with both landlord identity and macroeconomic conditions. Larger landlords are consistently less likely to set rents at multiples of 100 and more likely to use left-digit pricing, but these differences are especially pronounced during periods of rapid rent growth. This pattern reinforces our broader interpretation: behavioral pricing patterns are not fixed, but state-dependent, and larger landlords are both more flexible and more sophisticated in adapting pricing strategies to changing market conditions.

Figure 14 illustrates how the relationship between nominal rent changes and heuristic pricing varies by landlord size. Panel (A) plots the association between bunching at multiples of 100 and changes in annual rent prices, separately for small and large landlords. The non-monotonic pattern observed on average is driven primarily by small landlords. As rent growth deviates from its median value, the likelihood of bunching at 100 rises sharply for small landlords. This effect is particularly strong when rents increase above the median: we estimate a coefficient close to 1, implying that each 1 percentage point increase in rent inflation raises the likelihood of bunching at 100 by roughly 1 percentage point.

In contrast, large landlords display a much weaker pattern. When rents fall below the median, we estimate a similar increase in bunching at 100 as for small landlords, though the coefficient is not statistically significant. However, as rents rise above the median, large landlords show no increase in bunching. These results suggest that the overall non-monotonic relationship between bunching and rent inflation is primarily driven by smaller landlords. As nominal price changes grow larger, small landlords are more likely to revert to coarse pricing, consistent with cognitive frictions that impair optimization precision. For large landlords, evidence of behavioral responses is limited to downturns, where deflationary environments

may trigger modest increases in bunching. These findings show that cognitive frictions and behavioral pricing responses vary systematically with both market conditions and landlord sophistication.

Panels (B) and (C) show similar patterns when examining bunching at multiples of 50 and left-digit pricing, respectively. Panel (B) finds that small landlords again display strong non-monotonic behavior, with significant and asymmetric slope estimates: a negative coefficient of approximately -0.5 when rents fall below the median, and a positive coefficient of about 1.2 when rents rise above it. For large landlords, the slope is insignificant in the lower half of the rent distribution, and positive—but smaller—in the upper half, indicating modest increases in bunching at 50 when rent growth is high.

Panel (C) shows the relationship between rent inflation and left-digit pricing. Small landlords exhibit strong state dependence: left-digit pricing declines symmetrically as rents diverge from their median growth rate. For large landlords, we estimate a significant decline in left-digit pricing when rents fall, with a coefficient even larger than that for small landlords. However, as rents rise above the median, the slope for large landlords is much smaller and statistically insignificant. Overall, we find strong evidence that measures of sophisticated pricing strategies correlate more strongly with inflation for smaller landlords than larger landlords.

Figure 15 confirms these patterns using our alternative source of variation: the time between tenant turnovers. Among small landlords, longer gaps between leases—which require larger nominal resets—lead to more bunching at round numbers and less left-digit pricing. For large landlords, we estimate precise null effects. These results reinforce our interpretation: cognitive costs associated with larger price changes impair price-setting precision for smaller, less sophisticated landlords, while larger landlords remain relatively unaffected.

4 Conclusion

We use novel data on the near-universe of rental contracts in Berkeley, California, to document new facts about how rental prices adjust over the business cycle. We first document widespread use of heuristic pricing in rental markets. Landlords frequently set rents at round numbers, especially multiples of 50 and 100, and engage in substantial left-digit pricing. The prevalence of this bunching is strongly predicted by landlord size and corporate status. We interpret this as evidence of heterogeneity in pricing sophistication: larger landlords appear better able to set off-grid rents and strategically exploit left-digit bias, while the ability of

all landlords to fill units despite seemingly suboptimal pricing suggests a meaningful degree of market power.

We also find that bunching increases when rent inflation deviates from its median and when longer intervals pass between leases. These patterns point to a novel cognitive cost of inflation: when nominal changes are unusually large, landlords, especially smaller ones, face greater difficulty in setting prices precisely. These frictions are not evenly distributed across firms but disproportionately affect smaller, less sophisticated landlords.

Second, we provide strong evidence of state-dependent downward nominal rent rigidity. During expansions, we observe excess mass in the distribution of rent changes just above zero and missing mass just below, indicating resistance to price cuts. These patterns disappear during downturns, suggesting that the rigidity is not binding in bad times. Instead, it appears to reflect behavioral frictions that bind primarily in expansions, and that dissipate in downturns. These results caution against extrapolating from evidence on wage or price rigidity in normal times to conclude that such frictions bind during recessions.

We also provide new evidence on how landlords respond to variation in rental demand induced by the business cycle. Landlords respond to cyclical fluctuations by changing both the frequency and magnitude of price adjustments. In response to positive shocks, they are more likely to raise rents and by larger amounts, while becoming less likely to cut rents or reduce them as steeply. This starkly contrasts with the non-housing components of the cost of living, where only the frequency of price increases covaries with inflation ([Nakamura and Steinsson, 2008](#)). Our findings align more closely with a model in which landlords face Calvo-style price-setting frictions, as opposed to the menu-cost models that better explain pricing behavior in other sectors. Our results offer new moments that can inform the calibration of Calvo models, and suggest that behavioral frictions, such as a reliance on heuristics, play an important role in shaping observed pricing dynamics.

We also contribute new empirical evidence on how pricing responds to the seasonal cycle. Rental prices and lease volumes show seasonal patterns similar in magnitude to those observed in home purchases ([Ngai and Tenreyro, 2014](#)). While [Ngai and Tenreyro \(2014\)](#) attribute these dynamics to thick-market effects and variation in match quality, our findings suggest a different mechanism is at play. Because rental contracts are shorter and match-specific surplus is lower, the similarity in seasonal dynamics across rental and purchase markets points to more general constraints in pricing behavior.

Finally, we show that landlord responses to the business and seasonal cycle are highly heterogeneous. Larger landlords are more responsive across both extensive and intensive

margins: they are more likely to cut rents in recessions, raise them during booms, and increase them more steeply in hot markets. These findings suggest that the growing role of large landlords may have important implications for how shocks are transmitted through the housing market and for the distributional effects of macroeconomic volatility, though a full accounting of those welfare consequences is left for future work.

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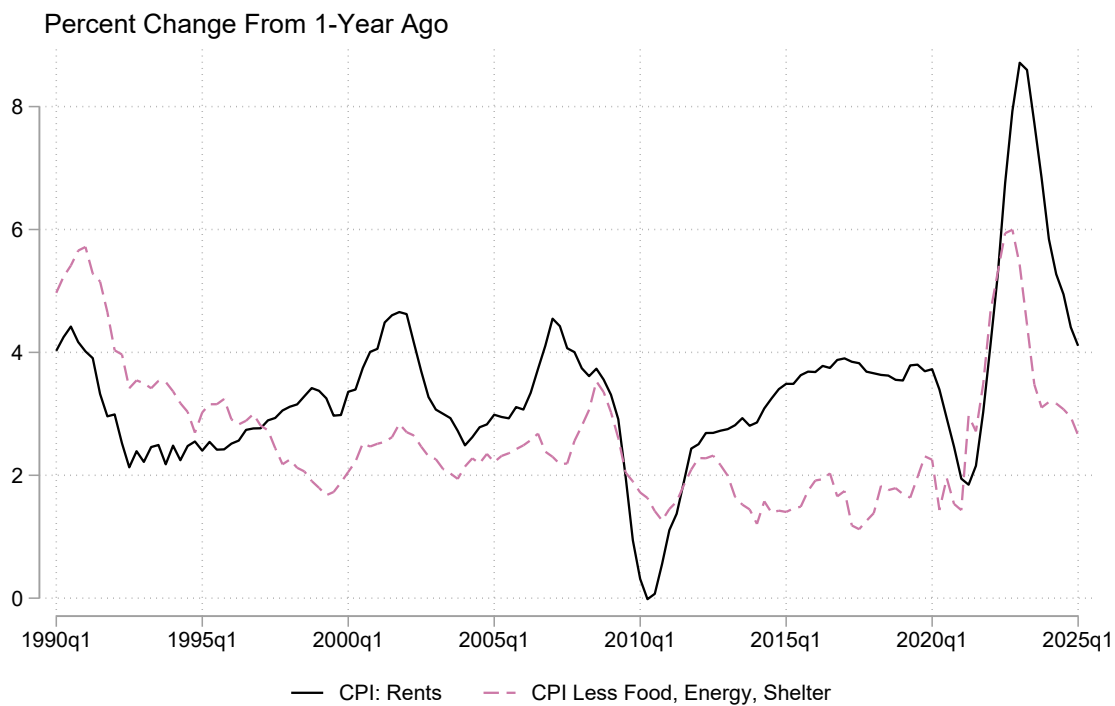
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5 Figures

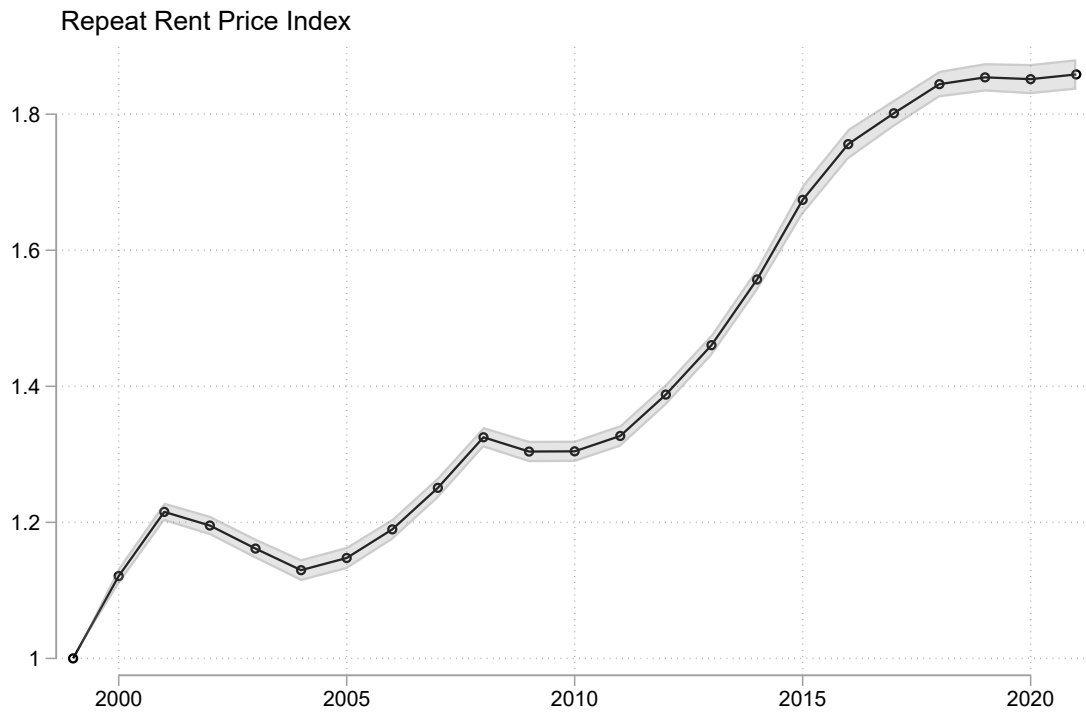
Figure 1: Time Series of Rent and Non-Housing Components of CPI



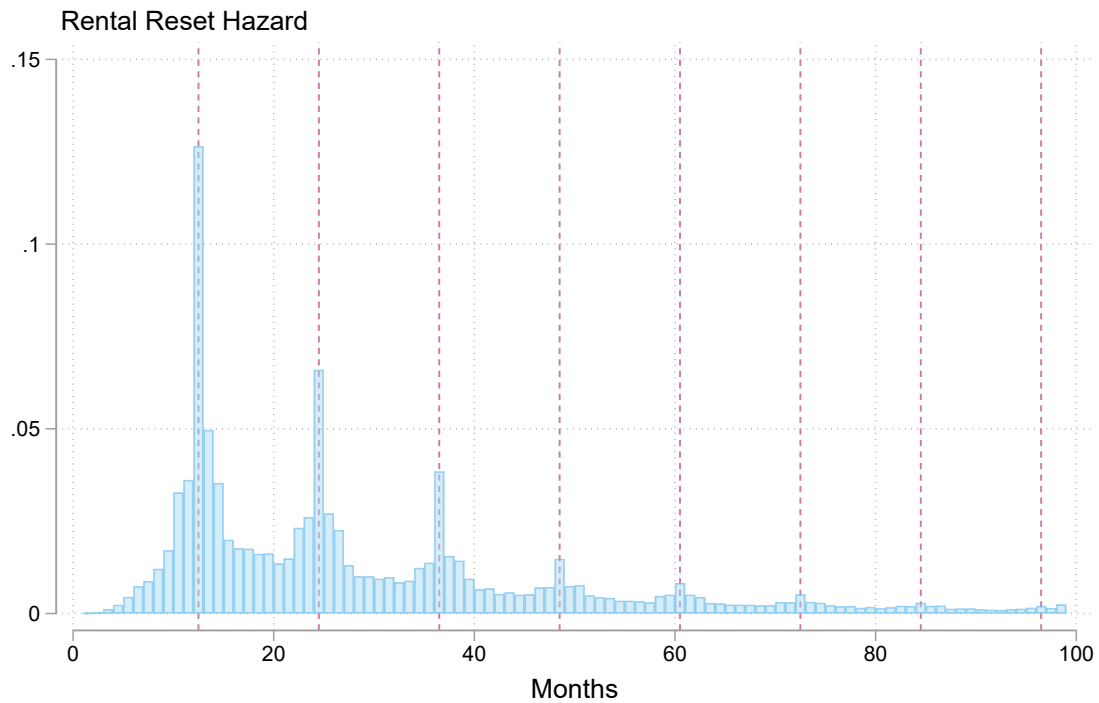
Notes: This figure plots the Consumer Price Index, excluding shelter, energy, and food in black. The line in pink shows the rental price component of the CPI.

Figure 2: Prices and Quantities in the Berkeley Rental Market

A. New-Tenant Repeat Rent Index

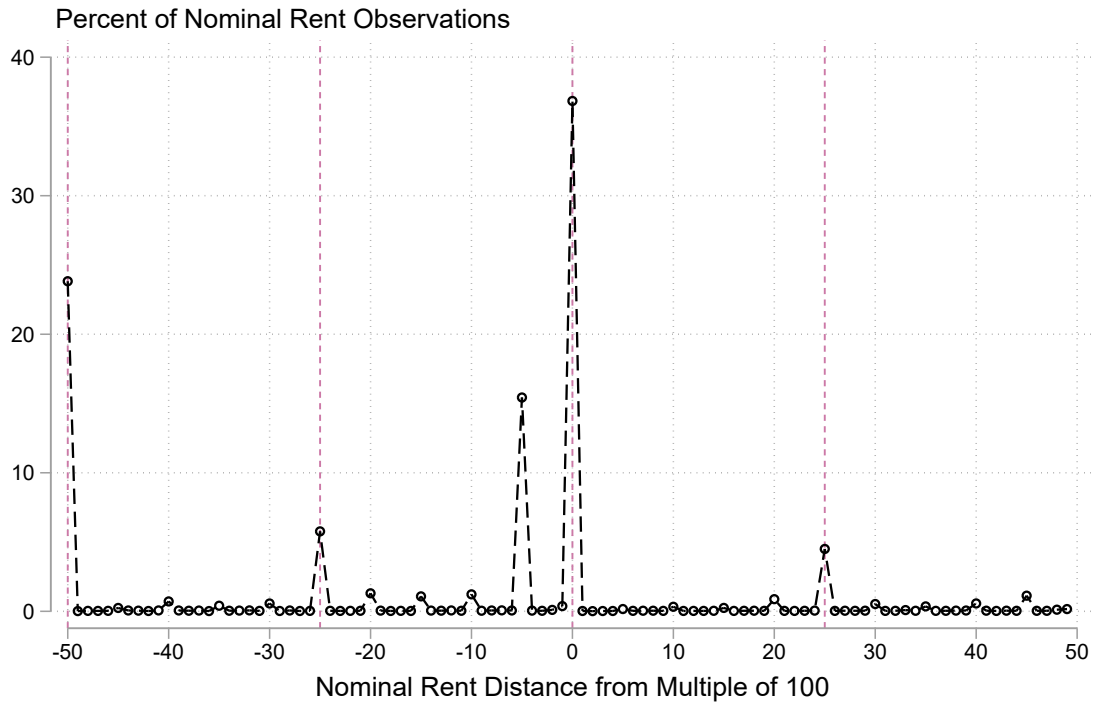


B. Hazard Rate of New Rental Contracts



Notes: Panel (A) shows the rent price index, extracted from a regression of new tenant rents on year and unit fixed effects. Panel (B) shows the distribution of time between new tenant rental contracts in Berkeley, California.

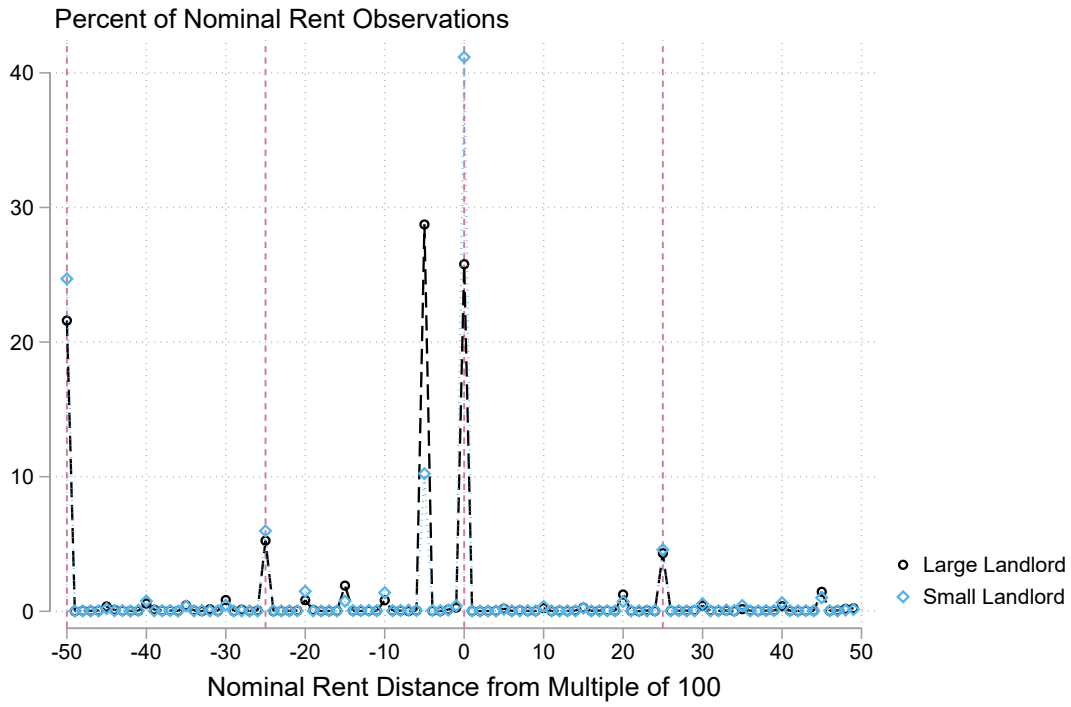
Figure 3: Heuristic Pricing in the Rental Market



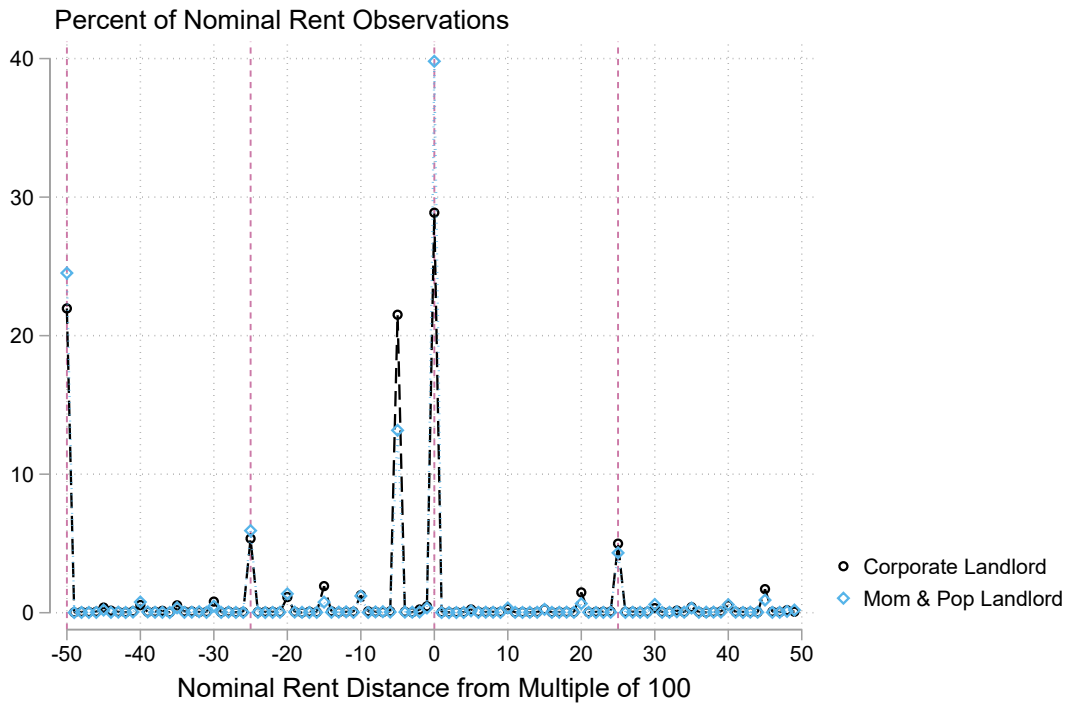
Notes: This figure shows the percent of all observations in the dataset with numbers based on their distance from the nearest multiple of \$100.

Figure 4: Heuristic Pricing by Landlord Type

A. Large vs. Small landlords



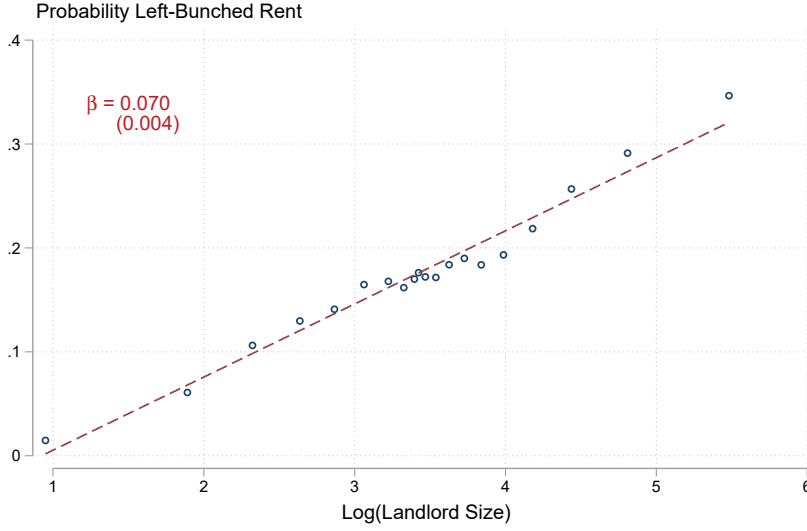
B. Corporate vs. Mom & Pop Landlords



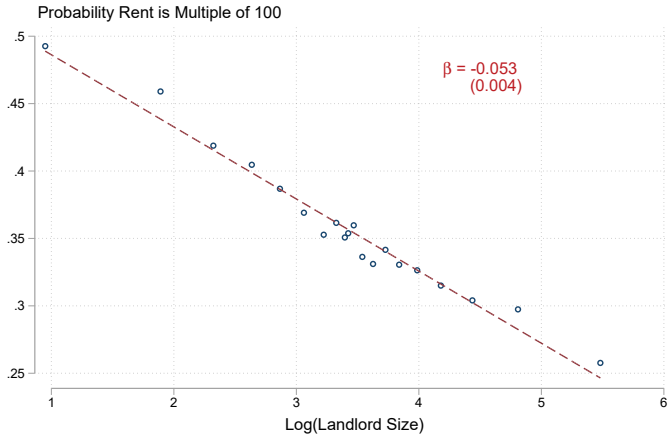
Notes: This figure shows the percent of all observations in the dataset with numbers based on their distance from the nearest multiple of \$100. In Panel (A) the results are shown separately for large landlords (defined as landlords with more than 75 units in Berkeley) and small landlords (all other observations). In Panel (B), we show the results separately for corporate and individual landlords.

Figure 5: Landlord Sophistication and Heuristic Pricing

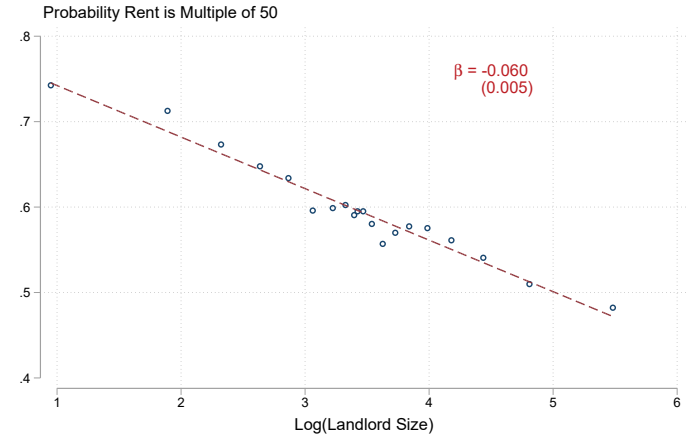
A. Left-Bunched Rents



B. Bunching at Multiple of 100

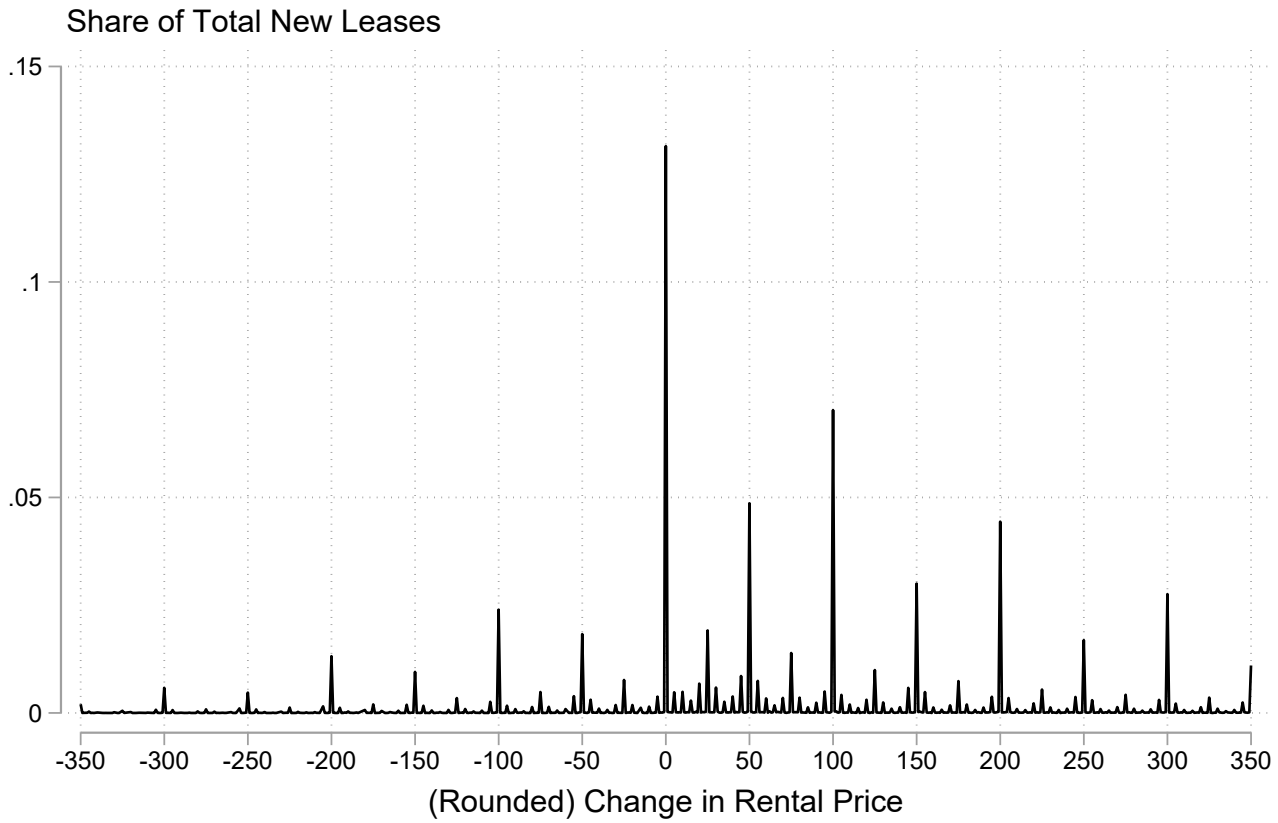


C. Bunching at Multiple of 50



Notes: These figures show binned scatter plots of different rent characteristics on $\text{Log}(\text{Landlord Size})$. All regressions include month-of-lease-start by month-of-last-lease-start and census-tract-by-year fixed effects. In Panel (A), the outcome variable is the probability of a left-bunched rent, defined as the probability that a rent observation is \$1 to \$9 less than a multiple of \$100. In Panel (B), the outcome variable is the probability the rent is an exact multiple of \$100. In Panel (C), the outcome variable is the probability the rent is an exact multiple of \$50. Point estimates and started errors clustered at the unit level are shown for each figure.

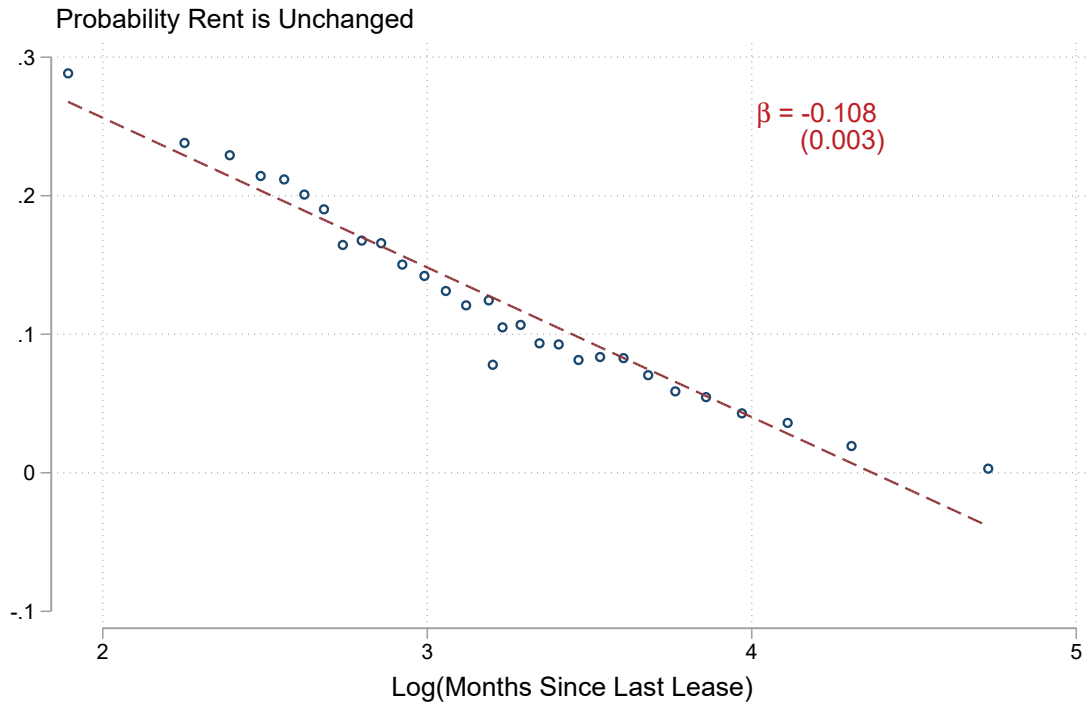
Figure 6: Nominal Dollar Change in New Tenant Rents



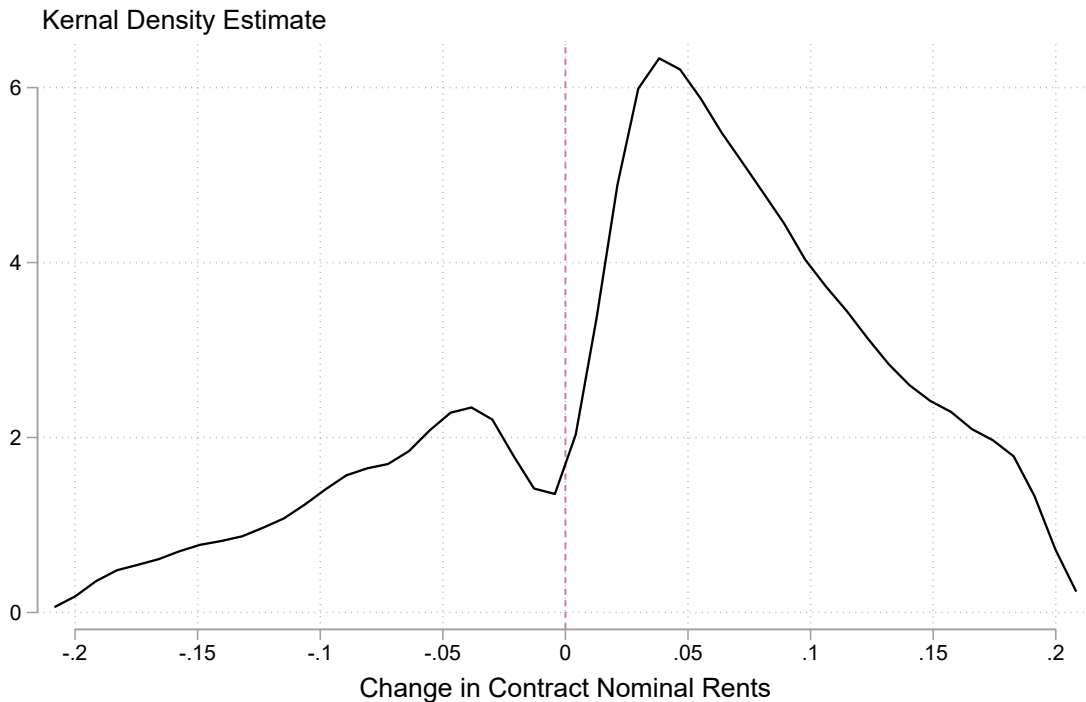
Notes: This figure shows a histogram of the share of new leases with each nominal rent change, rounded to the nearest dollar.

Figure 7: Evidence for Downward Nominal Rent Rigidity

A. Nominal Rent Rigidity Decreases with Time Between Leases



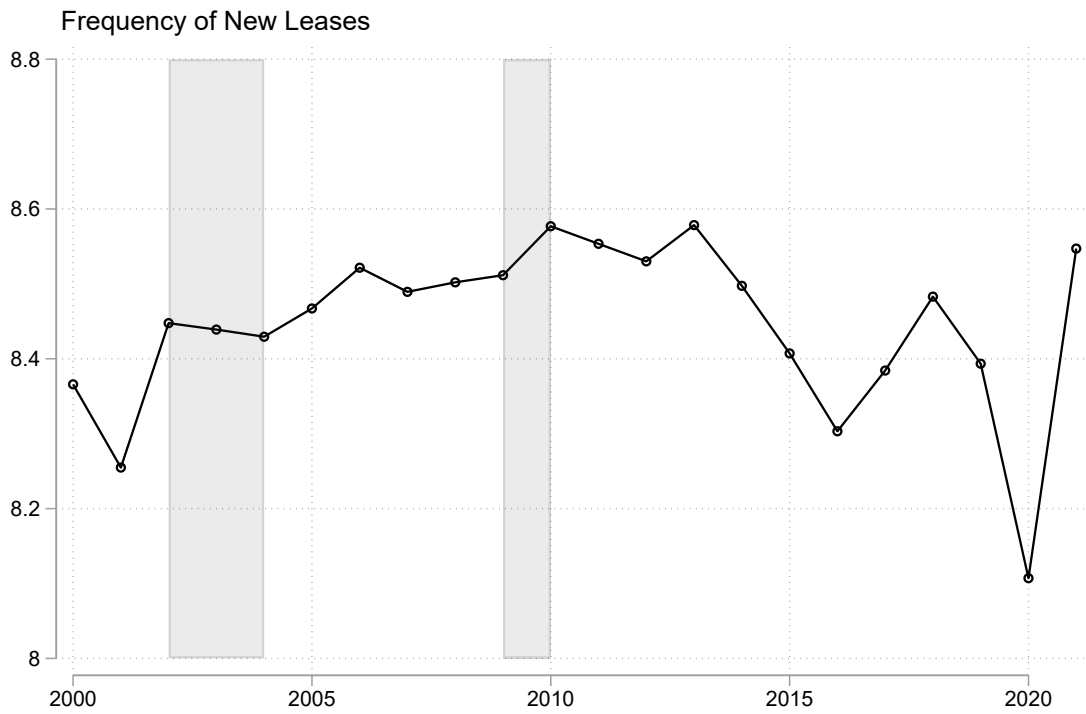
B. Kernel Density of Rent Changes (Excluding Zeros)



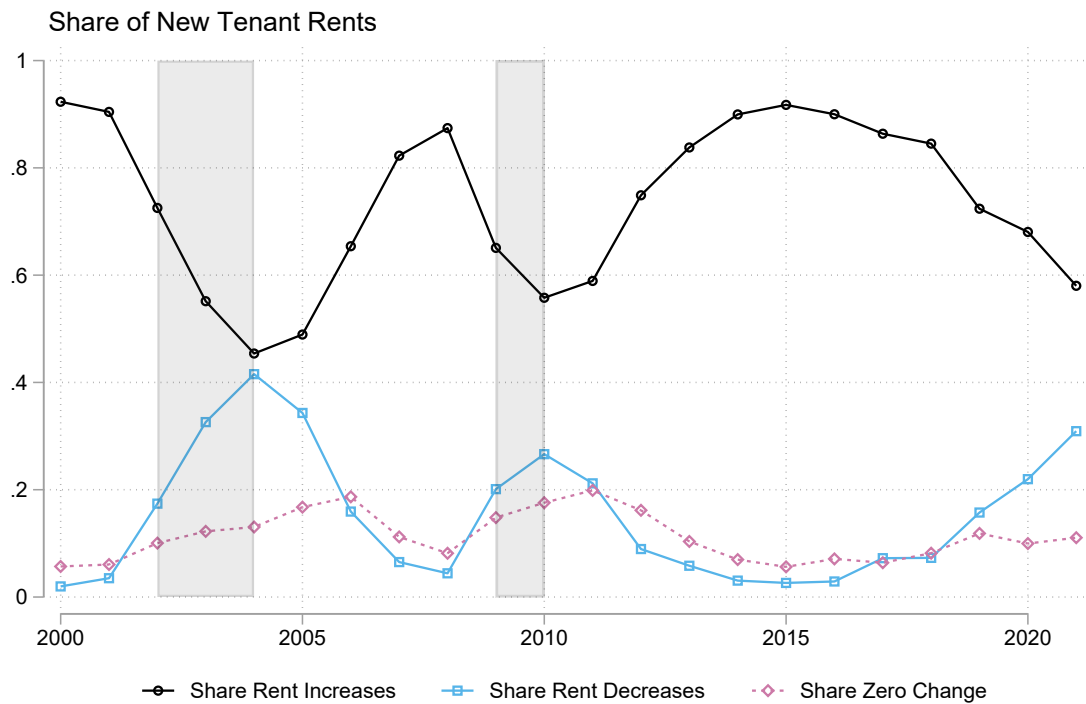
Notes: Panel (A) shows a binned scatter plots of a regression of the number of months since the previous lease on the probability the new lease comes with a zero rent change. The regression includes neighborhood-year fixed effects. Point estimates and standard errors clustered at the unit level are shown in parentheses. Panel (B) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that are less than 20 log points.

Figure 8: Frequency of Price Adjustments and the Business Cycle

A. Aggregate Mobility Does Not Vary Over Time

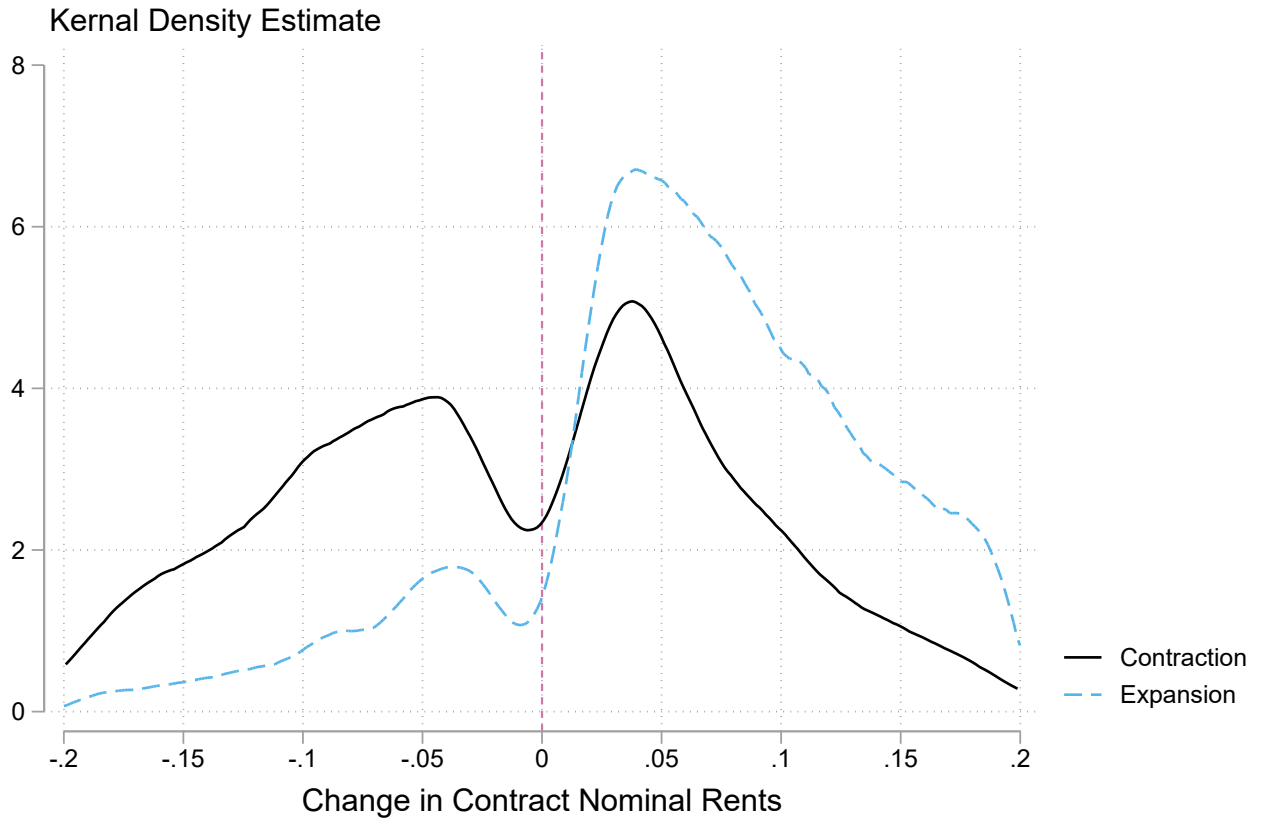


B. Frequency of Price Increases and Decreases



Notes: Panel (A) shows the time series of annual new leases in Berkeley, CA. Panel (B) shows the share of new tenant rents that come with a price increase (in black), a price decrease (in blue), or no change (in pink).

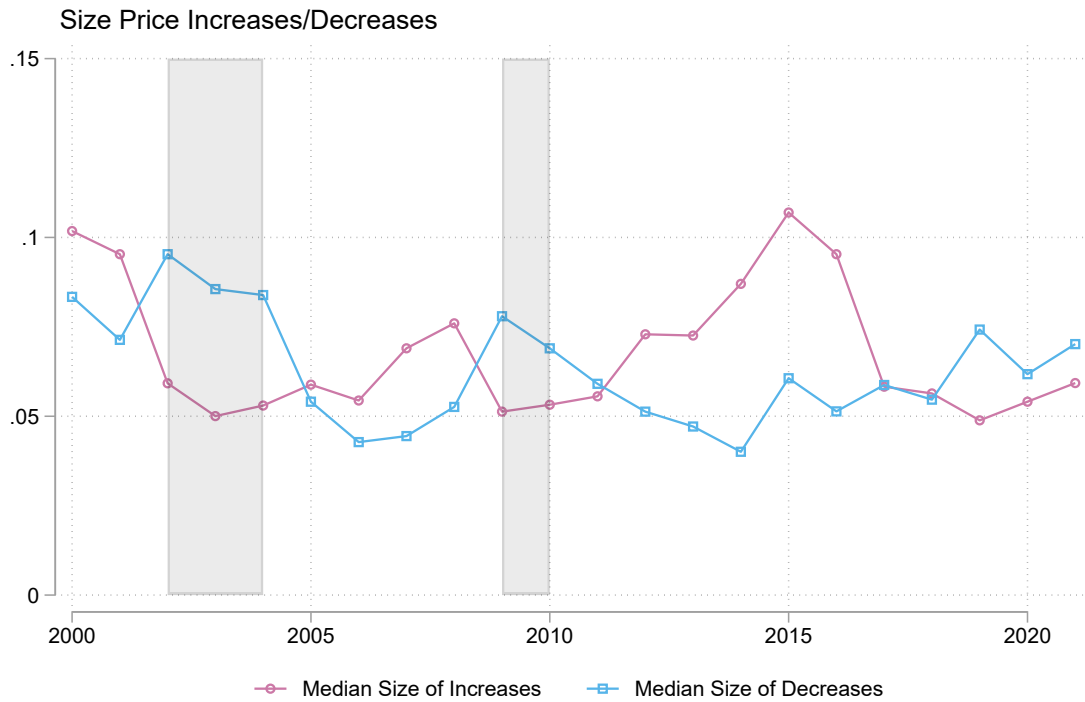
Figure 9: Downward Nominal Rent Rigidity is State-Dependent



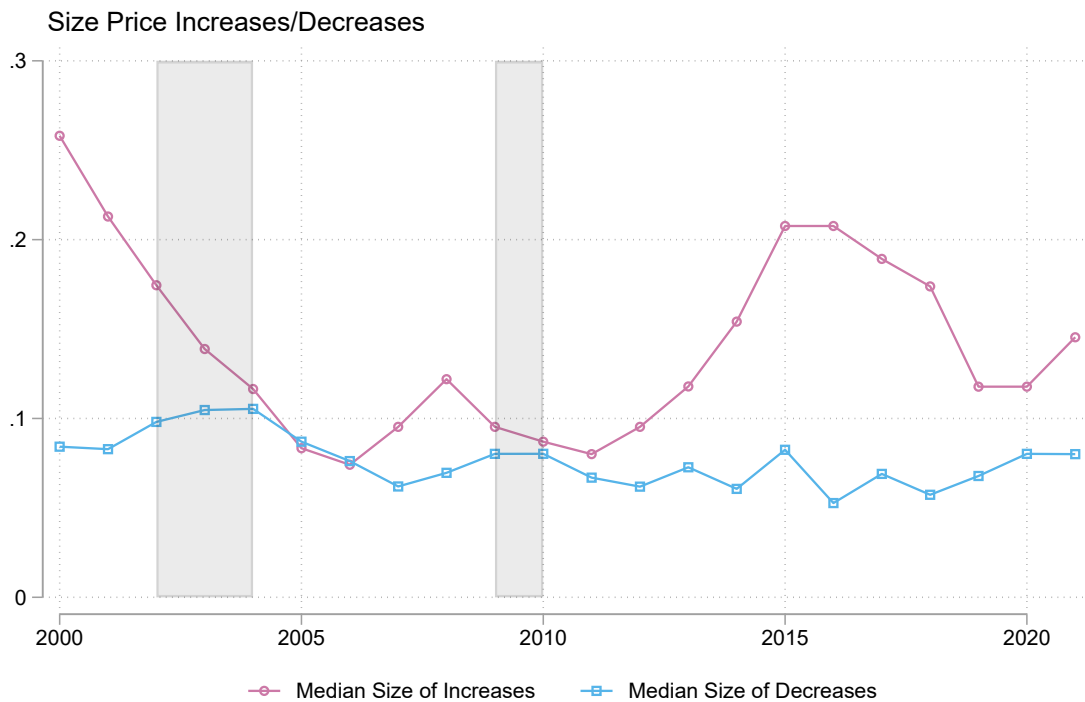
Notes: This figure shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that are less than 20 log points. In black, we estimate the kernel density for years when rents are falling. In blue, we show the kernel density plot for years when rents are increasing.

Figure 10: Magnitude of Price Adjustments and the Business Cycle

A. 1 Year or Less Since Previous Lease



B. All New Leases



Notes: This figure shows the annual time series of the median magnitude of price increases (in pink) and price decreases (in blue). Units are in log differences. Panel (A) shows new leases where the unit turns over within 12 months of the previous lease. Panel (B) shows all new leases.

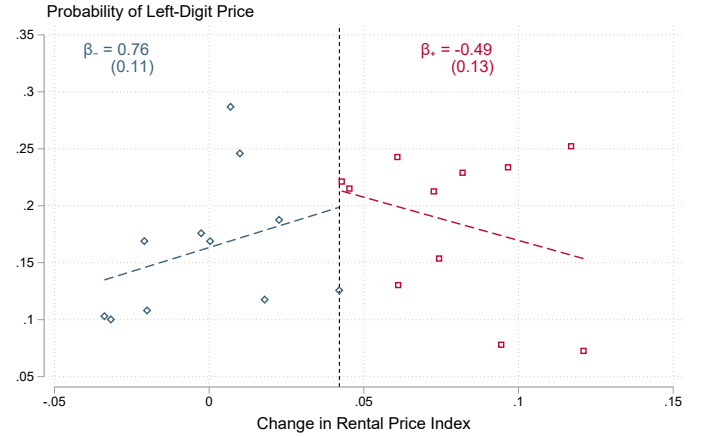
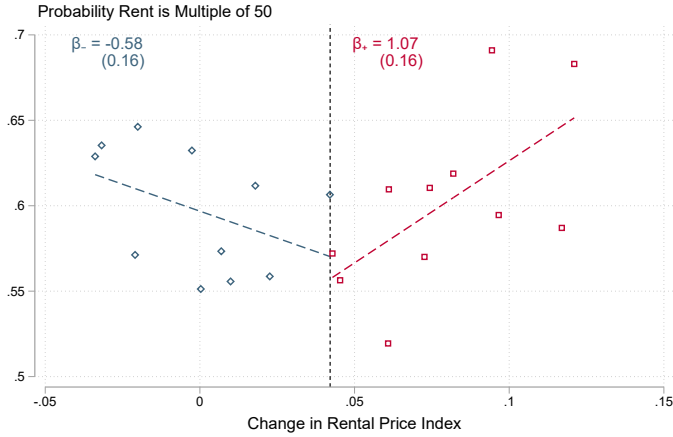
Figure 11: Heuristic Pricing and the Business Cycle

A. Bunching at Multiples of 100



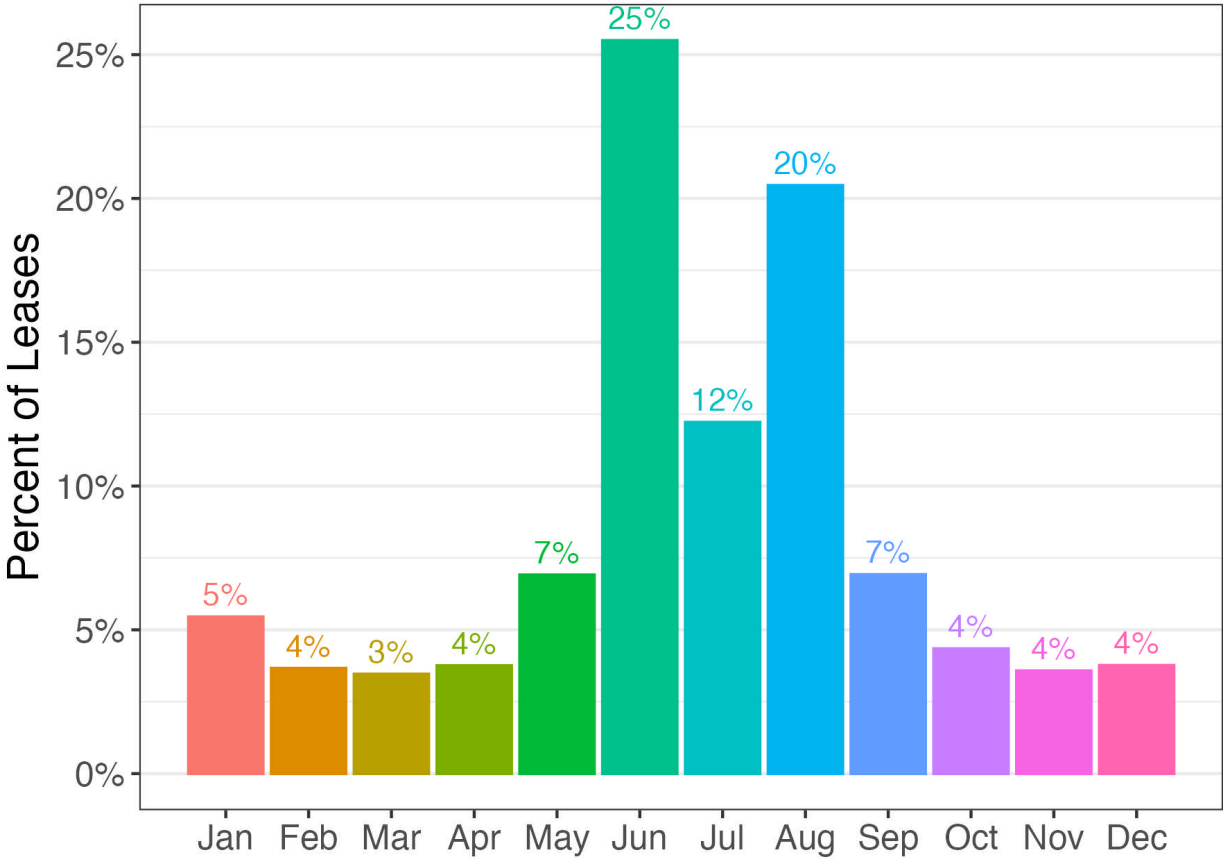
B. Bunching at Multiples of 50

C. Left-Digit Pricing



Notes: This figure shows binned scatter plots of the probability that a rent matches a particular characteristic on the annual change in the rental price index. We fit separate lines on either side of the 4.2%, which is approximately median price growth in our sample. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability the rent is a multiple of \$50. In Panel (C), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Outcome variables are residualized on time-since-last-lease fixed effects. Point estimates and standard errors clustered by unit are shown.

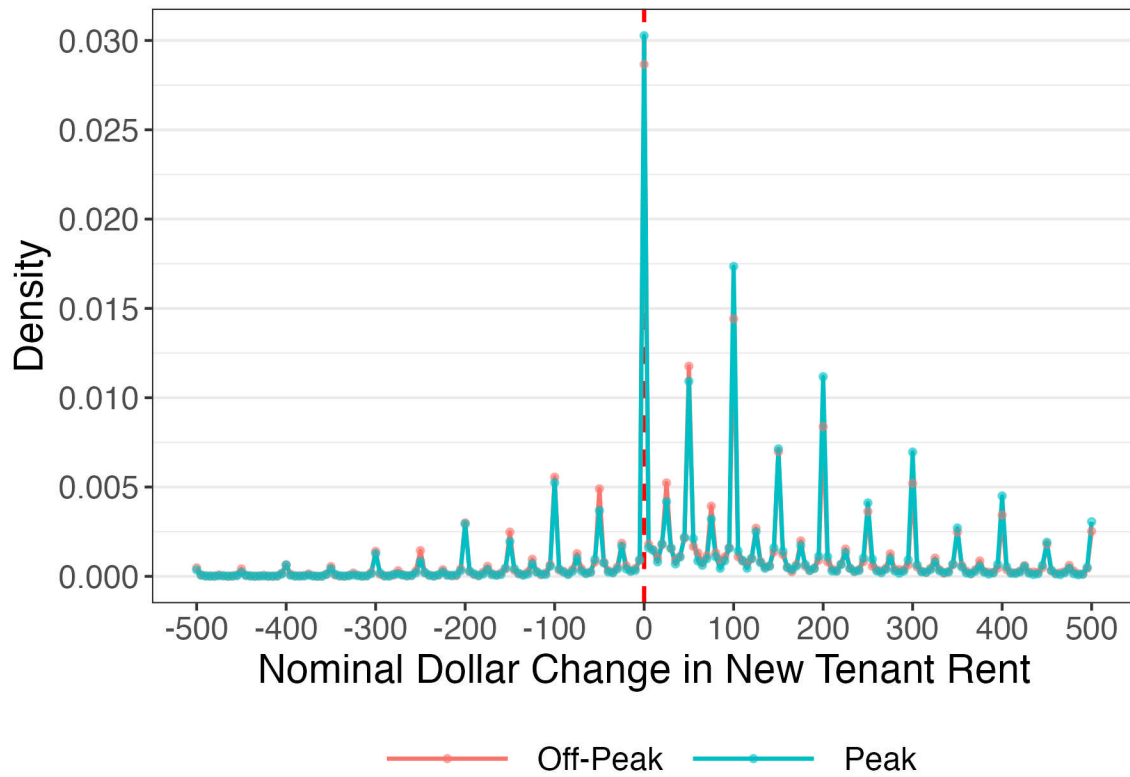
Figure 12: Seasonality in New Leases



Notes: This figure plots the percent of leases that start in each month.

Figure 13: Seasonal Price Variation

A. Peak vs. Off-Peak Price Change Distribution



B. Monthly Rental Price Premium

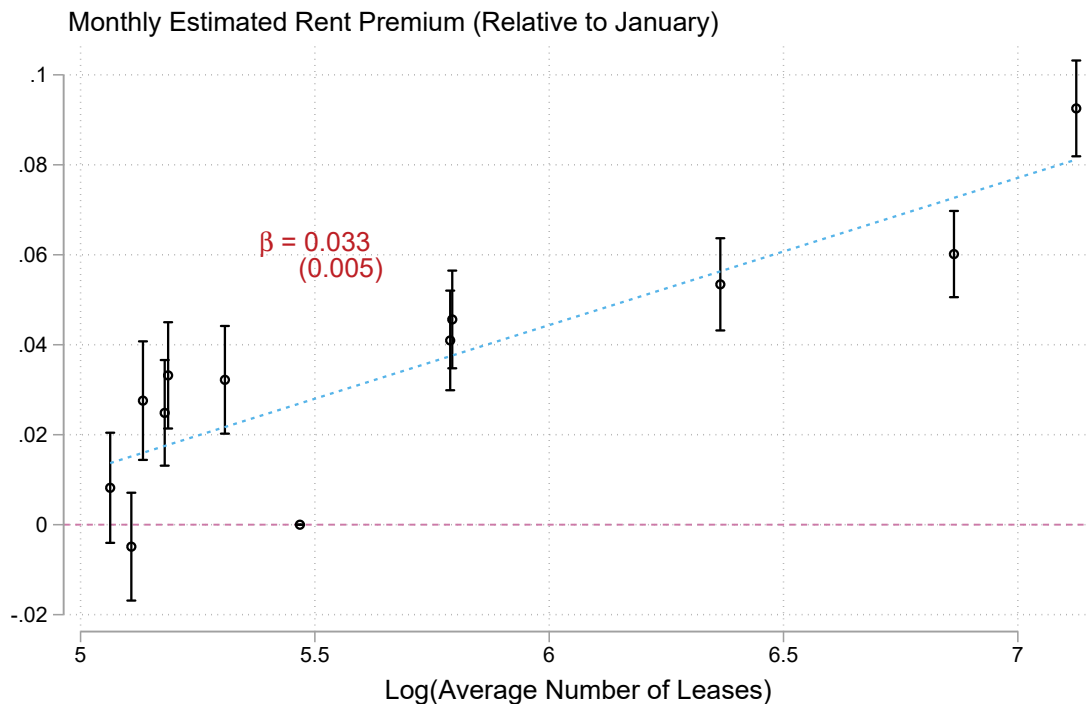
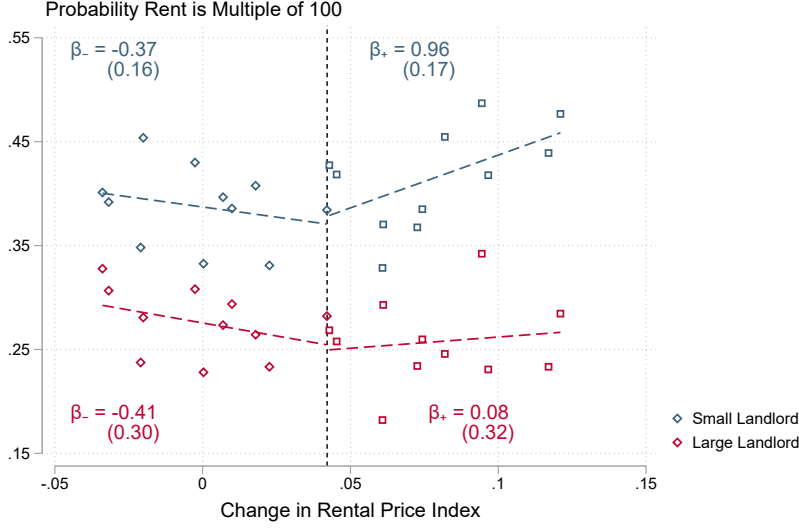


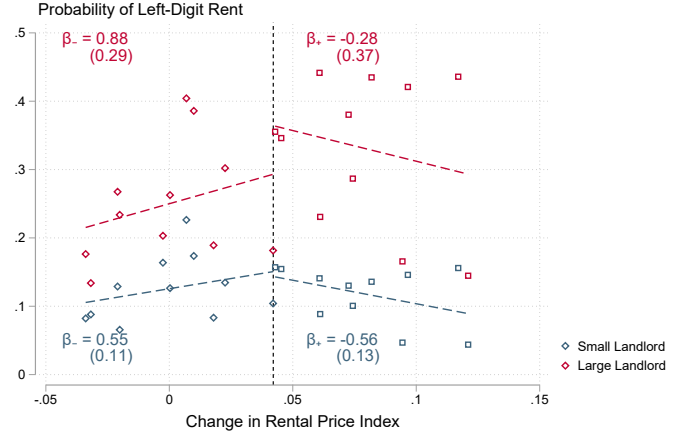
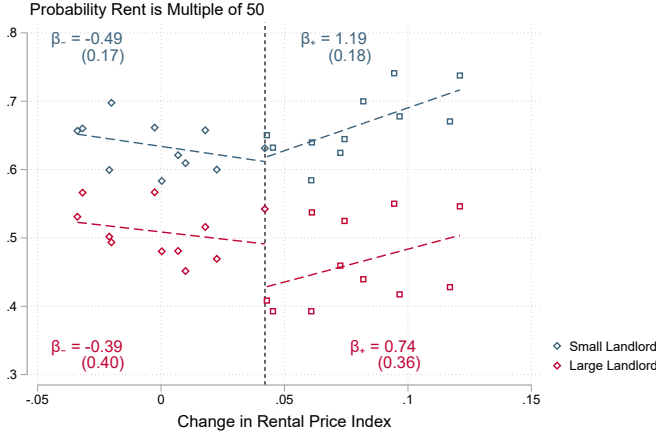
Figure 14: Heterogeneous Heuristic Pricing and the Business Cycle

A. Bunching at Multiples of 100



B. Bunching at Multiples of 50

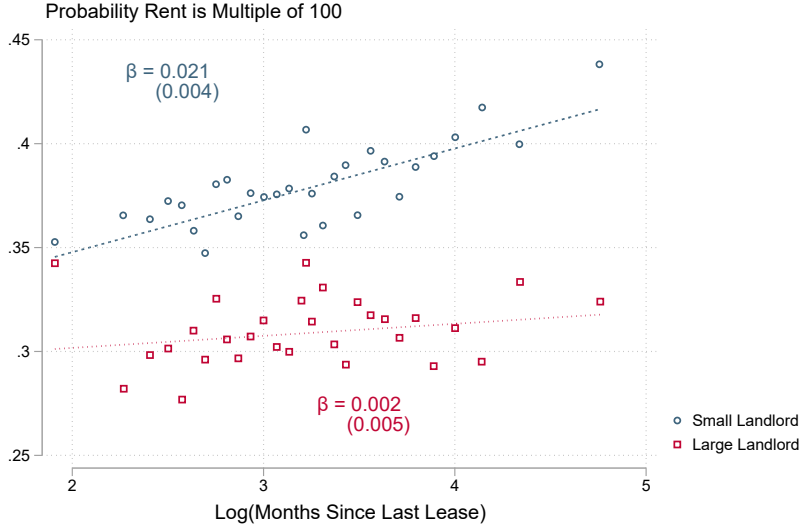
C. Left-Digit Pricing



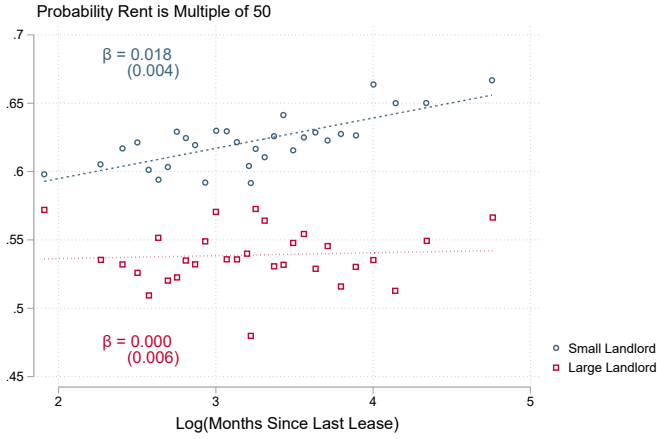
Notes: This figure shows binned scatter plots of the probability that a rent matches a particular characteristic on the annual change in the rental price index, separately for large (75 units in Berkeley) and small landlords. We fit separate lines on either side of the 4.2%, which is approximately median price growth in our sample. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability the rent is a multiple of \$50. In Panel (C), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Outcome variables are residualized on time-since-last-lease fixed effects. Point estimates and standard errors clustered by unit are shown.

Figure 15: Heterogeneous Heuristic Pricing and Time Since Last Lease

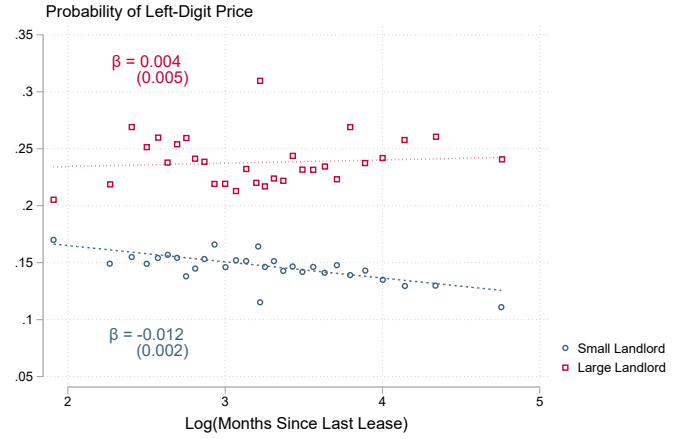
A. Bunching at Multiples of 100



B. Bunching at Multiples of 50



C. Left-Digit Pricing



Notes: This figure shows binned scatter plots of the probability that a rent matches a particular characteristic on the time since the previous lease, separately for large (≥ 75 units in Berkeley) and small landlords. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability the rent is a multiple of \$50. In Panel (C), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Outcome variables are residualized on census-tract-by-year fixed effects. Point estimates and standard errors clustered by unit are shown.

6 Tables

Table 1: Frequency of Price Adjustment

	Tot. Leases	Increases	Decreases	Unchanged
D. Rents	-0.24 (0.48)	3.53*** (0.69)	-19.1*** (1.48)	-5.66*** (1.36)
N	22	22	22	22
R^2	0.010	0.58	0.84	0.33

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Median Size of Price Adjustments

	≤ 1 Yr. Since Last Lease		All Leases	
	Increase	Decrease	Increase	Decrease
D. Rents	0.34*** (0.039)	-0.17* (0.080)	0.58** (0.20)	-0.16** (0.071)
N	22	22	22	22
R^2	0.79	0.27	0.29	0.27

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Rent Price Behavior in Peak vs. Off-Peak Months

	Log(New Leases)	Zero Spike	Price Cut	Big Increase	Left-Digit Price	Log(Rent)
1(Peak Month)	1.477*** (0.053)	-0.009*** (0.003)	-0.039*** (0.004)	0.026*** (0.003)	-0.000 (0.004)	0.044*** (0.002)
<i>N</i>	275	89815	89815	89815	89815	89815
R ²	0.789	0.273	0.347	0.476	0.441	0.823
Nbhood-YR FE	No	Yes	Yes	Yes	Yes	Yes
Unit-FE	No	Yes	Yes	Yes	Yes	Yes
Dep. Mean	5.65	0.13	0.18	0.28	0.19	7.34

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Landlord Size and Rental Pricing

	Rent Change is...				Rent is...	
	Zero Spike	Price Cut	Big Increase	Log Change	Multiple 100	Left-Digit Price
Landlord Size	-0.005** (0.002)	0.008*** (0.002)	0.008*** (0.003)	0.002*** (0.001)	-0.053*** (0.004)	0.070*** (0.004)
N	89494	89494	89494	89494	89494	89494
R^2	0.224	0.315	0.438	0.520	0.245	0.303
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.13	0.17	0.28	0.13	0.35	0.18

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Heterogeneous Rental Pricing Over the Business Cycle

	Rent Change is...				Rent is...	
	Price Increase	Price Cut	Big Increase	Log Change	Multiple 100	Left-Digit Price
Landlord Size	-0.015*** (0.003)	0.018*** (0.004)	0.004 (0.003)	-0.001 (0.001)	-0.045*** (0.005)	0.060*** (0.005)
x Tercile 2	0.015*** (0.005)	-0.010** (0.004)	0.001 (0.004)	0.002 (0.002)	-0.006 (0.005)	0.009** (0.004)
x Tercile 3	0.023*** (0.005)	-0.022*** (0.004)	0.014*** (0.004)	0.008*** (0.002)	-0.018*** (0.006)	0.028*** (0.005)
<i>N</i>	87269	87269	87269	87269	87269	87269
R ²	0.357	0.313	0.438	0.514	0.245	0.304
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.28	0.12	0.35	0.18

Standard errors in parentheses

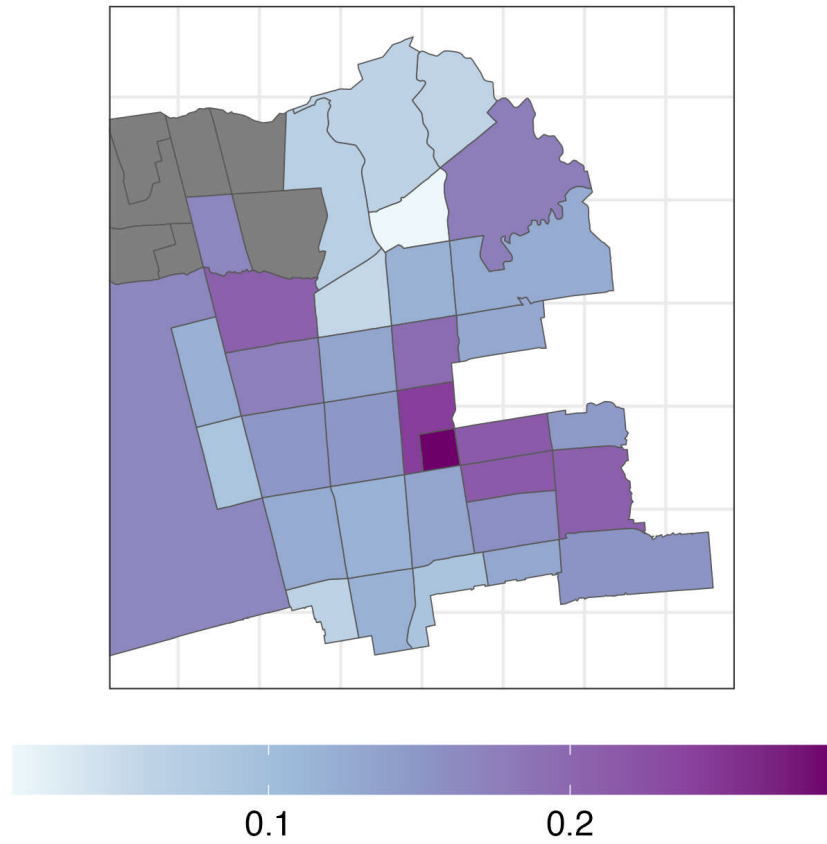
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Appendices Contents

- Appendix A1: Additional Figures
- Appendix A2: Additional Tables
- Appendix A3: Evidence from AHS

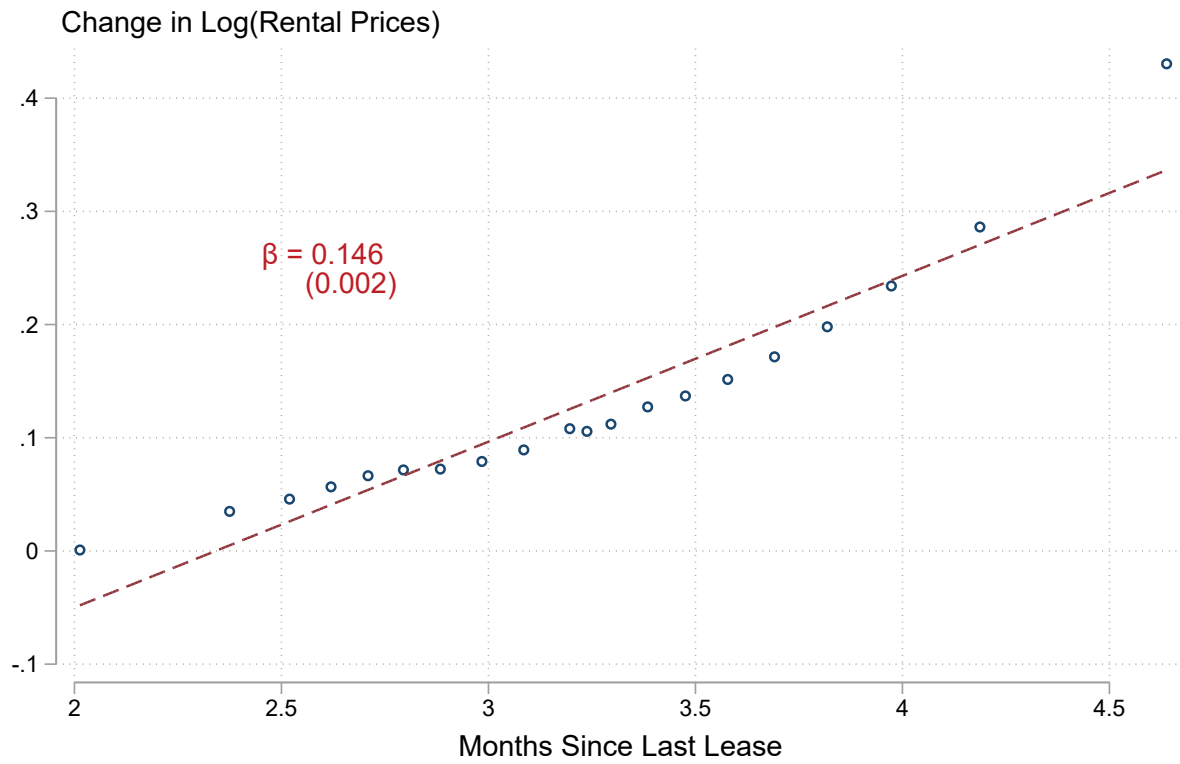
A1 Appendix Figures

Figure A1: Map of Left-Digit Pricing in Berkeley, California



Notes: This figure shows a census tract level map with the average level of left-digit pricing in each census tract in Berkeley.

Figure A2: Reset Prices Increase with the Time Between Leases



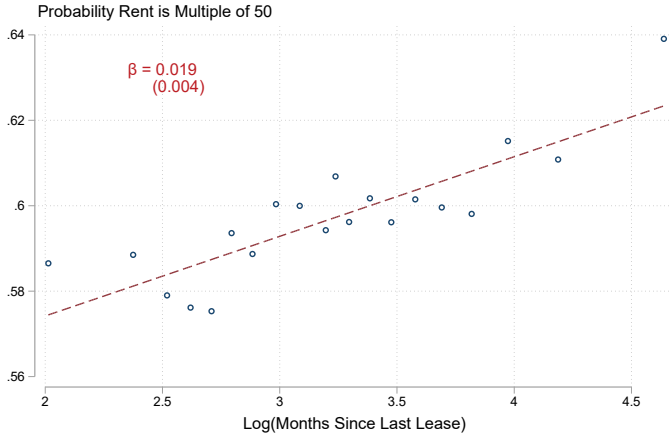
Notes: This figure shows binned scatter plots of the change in log rental prices on the number of months since the last lease (in logs). The regression includes census-tract-by-year fixed effects. Point estimates and standard errors clustered at the unit level are shown.

Figure A3: Heuristic Pricing and Time Between New Tenant Leases

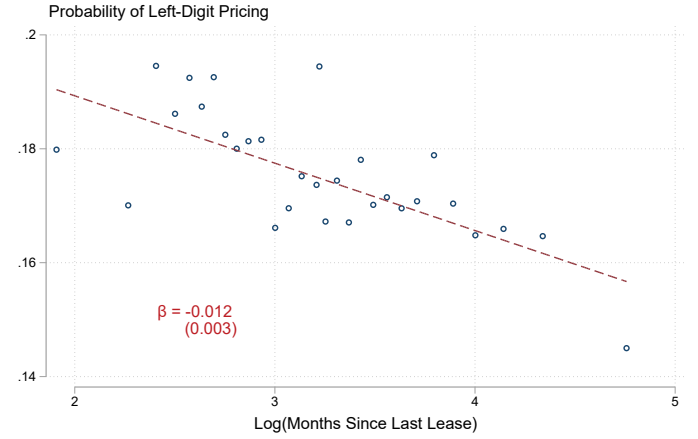
A. Bunching at Multiples of 100



B. Bunching at Multiples of 50



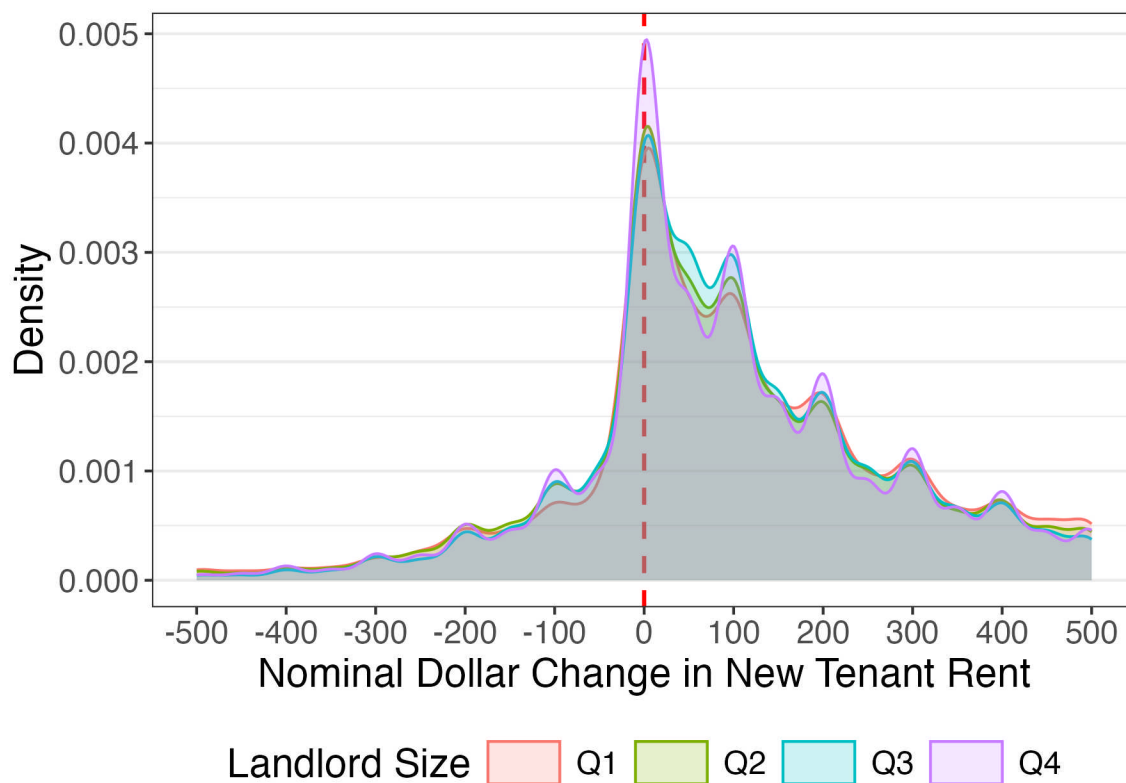
C. Left-Digit Pricing



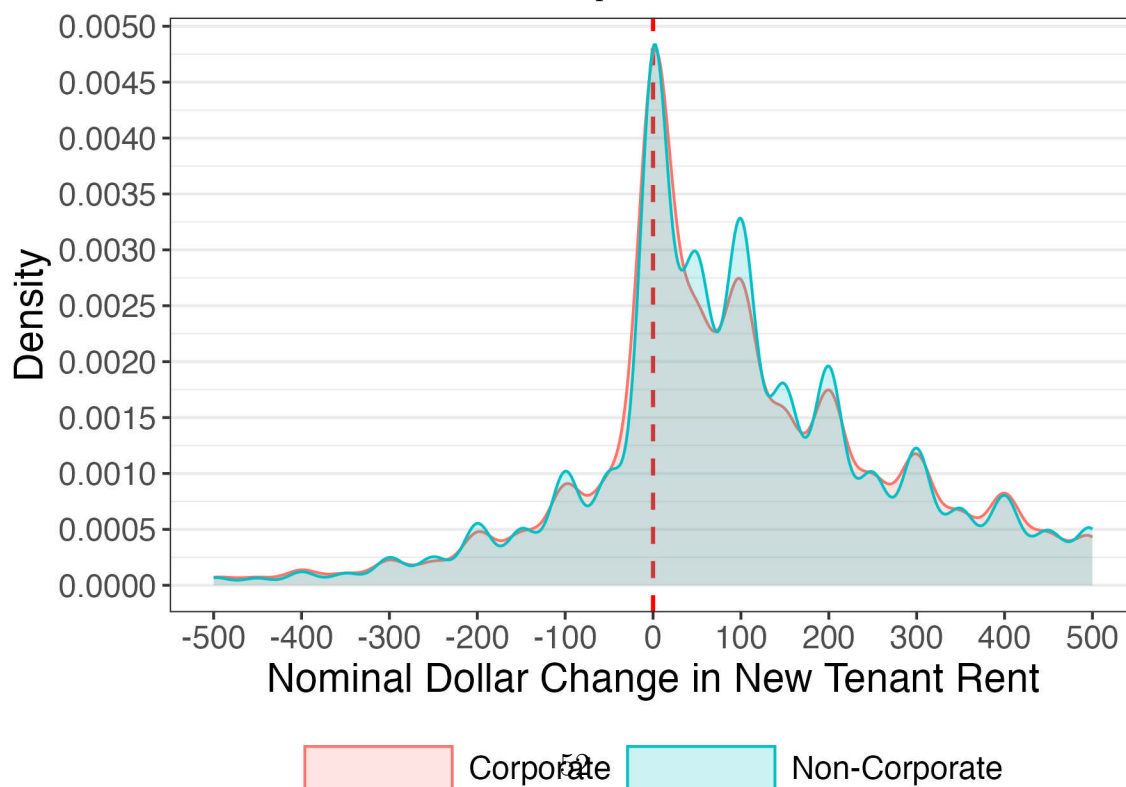
Notes: This figure shows binned scatter plots of the probability that a rent matches a particular characteristic on the time since the previous lease. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability the rent is a multiple of \$50. In Panel (C), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Outcome variables are residualized on census-tract-by-year fixed effects. Point estimates and standard errors clustered by unit are shown.

Figure A4: Unconditional Price Change Distribution by Landlord Characteristics

A. Landlord Size Quartile



B. Landlord Corporate Status



Notes: This figure shows kernel density plots of the nominal dollar change in new tenant rents. Panel (A) shows the estimates separately for different quartiles of the landlord size distribution. Panel (B) shows the estimates separately for corporate vs. non-corporate landlords.

A2 Appendix Tables

Table A1: Using Within-Unit Variation to Gauge How The Business Cycle Affects Rents

	Price Increase	Price Cut	Zero Spike	Big Increase	Big Decrease	D.Log(Rent)
D. Rent Index	0.034*** (0.001)	-0.027*** (0.001)	-0.007*** (0.001)	0.024*** (0.001)	-0.015*** (0.000)	0.013*** (0.000)
<i>N</i>	89605	89605	89605	89605	89605	89605
R ²	0.232	0.183	0.155	0.307	0.133	0.385
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.28	0.09	0.12

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A2: Corporate Landlords and Rental Pricing

	Rent Change is...				Rent is...	
	Zero Spike	Price Cut	Big Increase	Log Change	Multiple 100	Left-Digit Price
Corp. Landlord	-0.014* (0.008)	0.001 (0.007)	0.022** (0.009)	0.011*** (0.003)	-0.089*** (0.018)	0.041** (0.018)
<i>N</i>	89494	89494	89494	89494	89494	89494
R ²	0.224	0.315	0.438	0.520	0.236	0.270
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.13	0.17	0.28	0.13	0.35	0.18

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Very Large Landlords and Rental Pricing

	Rent Change is...				Rent is...	
	Zero Spike	Price Cut	Big Increase	Log Change	Multiple 100	Left-Digit Price
Big Landlord	0.005 (0.011)	0.010 (0.007)	0.027*** (0.010)	0.007** (0.003)	-0.139*** (0.019)	0.179*** (0.018)
N	89494	89494	89494	89494	89494	89494
R^2	0.224	0.315	0.438	0.520	0.241	0.292
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.13	0.17	0.28	0.13	0.35	0.18

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Heterogeneous Responses to Seasonal Variation

	Zero Spike	Price Cut	Big Increase	Left-Digit Price	Log(Rent)
1(Peak Month)	0.002 (0.007)	-0.041*** (0.008)	0.029*** (0.009)	-0.007 (0.008)	0.025*** (0.006)
x Landlord Size	-0.003 (0.002)	0.001 (0.002)	-0.001 (0.002)	0.002 (0.003)	0.005*** (0.002)
N	89815	89815	89815	89815	89815
R^2	0.273	0.347	0.476	0.443	0.823
Nbhood-YR FE	Yes	Yes	Yes	Yes	Yes
Unit-FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.13	0.18	0.28	0.19	7.34

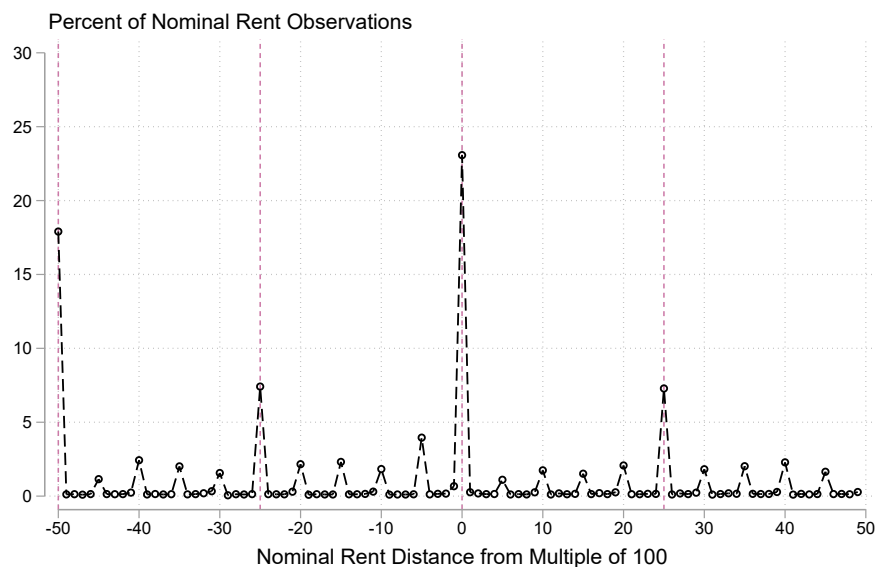
Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

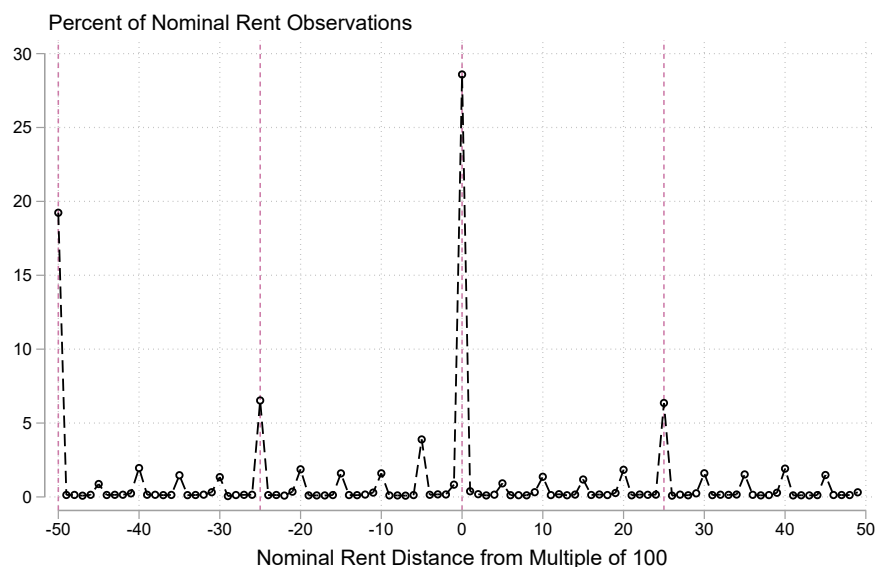
A3 Evidence from the American Housing Survey

Figure A5: Heuristic Pricing in the Broader American Rental Market

A. 1985 - 2013 Sample



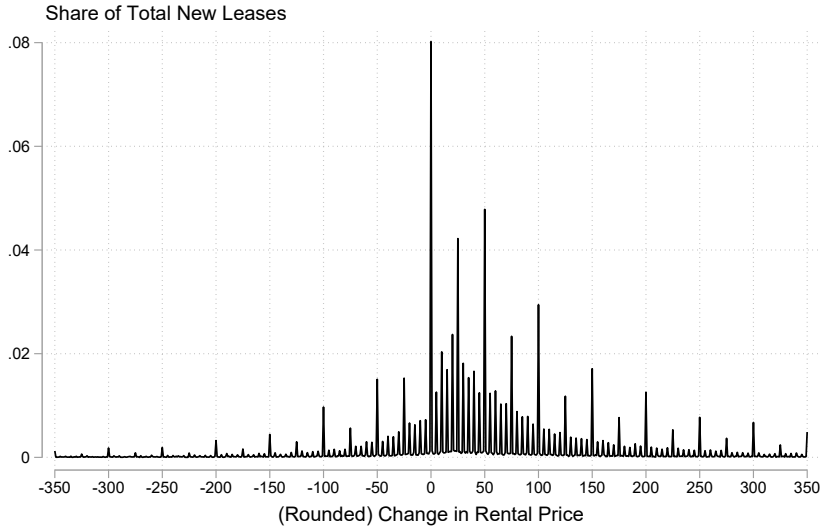
B. 1999 - 2013 Sample



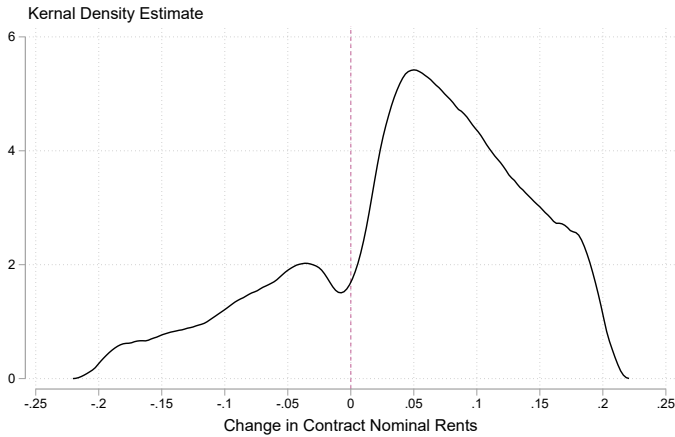
Notes: This figure shows the percent of all observations in the AHS dataset with numbers based on their distance from the nearest multiple of \$100. Panel (A) shows the results for the period between 1985 and 2013, while Panel (B) shows the results for the period between 1999 and 2013. All counts are weighted using their sampling weights.

Figure A6: Downward Nominal Rent Rigidity in the AHS

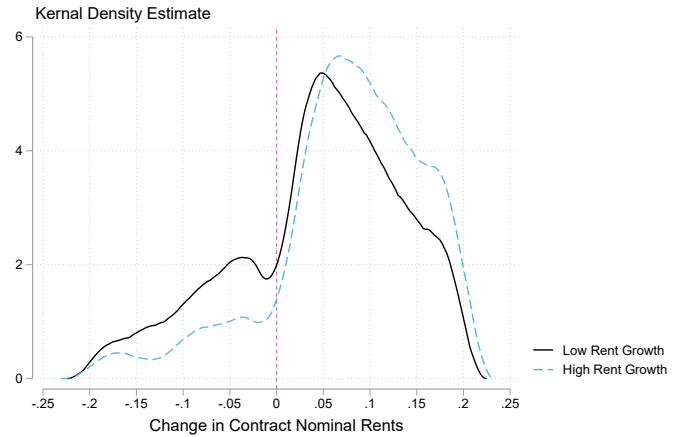
A. Overall Bunching at Zero Rent Changes



B. Kernel Density (excl. Zeros)



C. Kernel Densities by Rent Growth (excl. Zeros)



Notes: Panel (A) shows the histogram of the nominal dollar change in rents, rounded to the nearest dollar, in the AHS. We only include observations where the household differs from the household interviewed in the previous survey wave. Panel (B) shows the kernel density plots of the log change in contract rents, excluding zeros and dropping observations where the rent change is large at 20 log points. Panel (C) shows the same kernel density plots, but separately for low and high rent growth periods.

Table A5: Within Unit Responses to the Business Cycle

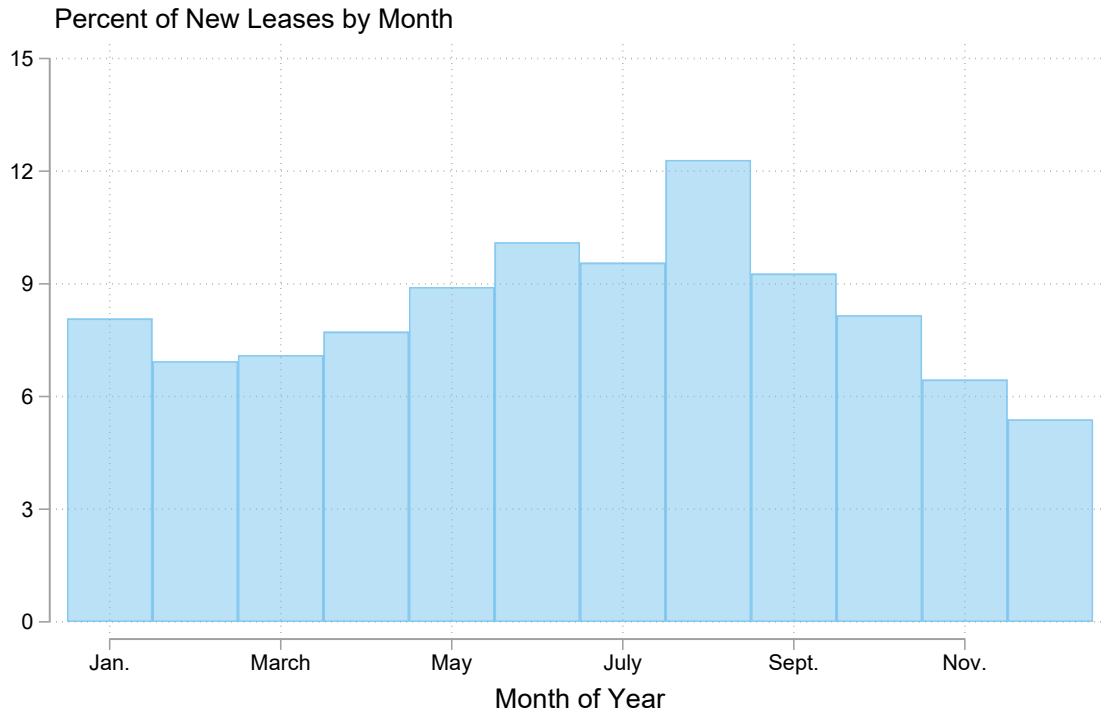
	Price Increase	Price Cut	Zero Spike	Big Increase	Big Decrease	D.Log(Rent)
D. Rent Index	0.039*** (0.003)	-0.032*** (0.003)	-0.007*** (0.002)	0.015*** (0.003)	-0.023*** (0.002)	0.021*** (0.004)
<i>N</i>	50356	50356	50356	50356	50356	50356
R ²	0.208	0.218	0.274	0.263	0.247	0.146
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.70	0.21	0.08	0.27	0.12	0.12

Standard errors in parentheses

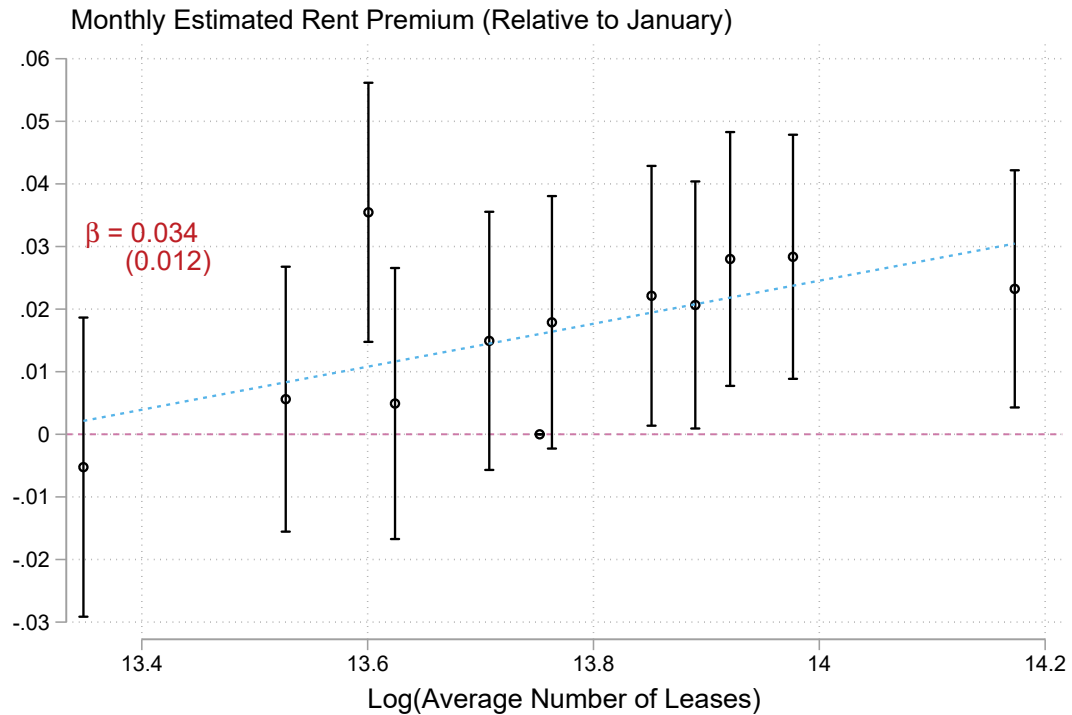
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A7: Seasonal Variation in the Broader U.S. Rental Market

A. Share of New Leases by Month



B. Elasticity of Rent Premia to Seasonal Leases



Notes: Panel (A) shows the share of leases that begin in a particular month. Panel (B) shows scatter plot of the estimated rent premia on the average number of leases per month (in logs). Monthly rent premia are estimated from a regression of log rents on month fixed effects, controlling for city-by-year and unit fixed effects. Estimated rent premia and standard errors are shown, as is the regression coefficient from a regression of the log(average number of leases) on the rent premia. Point estimates and robust standard errors for that regression are shown.

Table A6: Within Unit Responses to the Seasonal Cycle

	Log(New Leases)	Price Increase	Zero Spike	Price Cut	Big Increase	Log(Rent)
1(Peak Month)	0.350*** (0.034)	0.018*** (0.005)	-0.005* (0.003)	-0.013*** (0.005)	0.021*** (0.005)	0.012** (0.005)
<i>N</i>	168	45068	45068	45068	45068	45068
R ²	0.518	0.307	0.334	0.308	0.363	0.703
SMSA-Year FE	No	Yes	Yes	Yes	Yes	Yes
Unit FE	No	Yes	Yes	Yes	Yes	Yes
Dep. Mean	13.54	0.08	0.08	0.22	0.28	6.30

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$